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LENS DESIGN FOR LOW PARALLAX PANORAMIC CAMERA SYSTEMS

Abstract

A lens form for a low parallax imaging device includes a plurality of imaging lens elements arranged to capture adjacent fields of view. The imaging lens elements include a plurality of lens element groups for directing incident light to an imaging plane. The lens elements control a distance between an entrance pupil located behind the image plane and a low parallax point to reduce parallax errors between adjacent of the imaging lens elements.

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Background/Summary

CROSS REFERENCE TO RELATED APPLICATIONS [0001] This application is a Continuation of U.S. Non-provisional application Ser. No. 17/622,463, filed Dec. 23, 2021, titled, “LENS DESIGN FOR LOW PARALLAX PANORAMIC CAMERA SYSTEMS,” which is a 35 U.S.C. § 371 National Stage Entry of, and claims priority to, International Application No.

PCT/US2020/039200, filed Jun. 23, 2020, which claims benefit of priority of: U.S. Provisional Patent Application Ser. No. 62/865,741, filed Jun. 24, 2019, titled “Opto-Mechanics of Panoramic Capture Devices with Abutting Cameras;” U.S. Provisional Patent Application Ser. No. 62/952,973, filed Dec. 23, 2019, titled “Opto-Mechanics of Panoramic Capture Devices with Abutting Cameras;” U.S. Provisional Patent Application Ser. No. 62/952,983, filed Dec. 23, 2019, titled “Multi-camera Panoramic Image Capture Devices with a Faceted Dome;” and U.S. Provisional Patent Application Ser. No. 62/972,532, filed Feb. 10, 2020, titled “Integrated Depth Sensing and Panoramic Camera System,” the entirety of each of which is incorporated herein by reference.

TECHNICAL FIELD

[0002] The present disclosure relates to panoramic low-parallax multi-camera capture devices having a plurality of adjacent and abutting polygonal cameras. The disclosure also relates to the lens design of cameras that capture incident light from a polygonal shaped field of view to form a polygonal shaped image.

BACKGROUND

[0003] Panoramic cameras have substantial value because of their ability to simultaneously capture wide field of view images. The earliest such example is the fisheye lens, which is an ultra-wide-angle lens that produces strong visual distortion while capturing a wide panoramic or hemispherical image. While the field of view (FOV) of a fisheye lens is usually between 100 and 180 degrees, the approach has been extended to yet larger angles, including into the 220-270° range, as provided by Y. Shimizu in U.S. Pat. No. 3,524,697. As an alternative, there are mirror or reflective based cameras that capture annular panoramic images, such as the system suggested by P. Greguss in U.S. Pat. No. 4,930,864. While these technologies have continued to evolve, it is difficult for them to provide a full hemispheric or spherical image with the resolution and image quality that modern applications are now seeking.

[0004] As another alternative, panoramic multi-camera devices, with a plurality of cameras arranged around a sphere or a circumference of a sphere, are becoming increasingly common. However, in most of these systems, including those described in U.S. Pat. Nos. 9,451,162 and 9,911,454, both to A. Van Hoff et al., of Jaunt Inc., the plurality of cameras are sparsely populating the outer surface of the device. In order to capture complete 360-degree panoramic images, including for the gaps or seams between the adjacent individual cameras, the cameras then have widened FOVs that overlap one to another. In some cases, as much as 50% of a camera's FOV or resolution may be used for camera to camera overlap, which also creates substantial parallax

differences between the captured images. Parallax is the visual perception that the position or direction of an object appears to be different when viewed from different positions. Then in the subsequent image processing, the excess image overlap and parallax differences both complicate and significantly slow the efforts to properly combine, tile or stitch, and synthesize acceptable images from the images captured by adjacent cameras.

[0005] There are also panoramic multi-camera devices in which a plurality of cameras is arranged around a sphere or a circumference of a sphere, such that adjacent cameras are abutting along a part or the whole of adjacent edges. As an example, U.S. Pat. No. 7,515,177 by K. Yoshikawa depicts an imaging device with a multitude of adjacent image pickup units (cameras). Images are collected from cameras having overlapping fields of view, so as to compensate for mechanical errors.

[0006] There remain opportunities to improve the optical designs and function of imaging lens systems that can be used in low parallax panoramic multi-camera devices. The potential optical improvements could also have direct and indirect benefit or synergy for the opto-mechanical designs of the individual camera lens systems and the overall device, particularly as related to issues at or near the seams between adjacent cameras.

Description

BRIEF DESCRIPTION OF DRAWINGS

[0007] FIG. 1 depicts a 3D view of a portion of a multi-camera capture device, and specifically two adjacent cameras thereof.

[0008] FIGS. 2A and 2B depict portions of camera lens assemblies in cross-section, including lens elements and ray paths.

[0009] FIG. 3 depicts a cross-sectional view of a portion of a standard multi-camera capture device showing FOV overlap, Fields of view, overlap, seams, and blind regions.

[0010] FIG. 4 depicts two polyhedron shapes, that of a regular dodecahedron and a truncated icosahedron, to which a multi-camera capture device can be designed and fabricated.

[0011] FIGS. 5A and FIG. 5B depict the optical geometry for fields of view for adjacent hexagonal and pentagonal lenses, as can occur with a device having the geometry of a truncated icosahedron. FIG. 5B depicts an expanded area of FIG. 5A with greater detail.

[0012] FIG. 5C depicts an example of a low parallax (LP) volume located near both a paraxial NP point or entrance pupil and a device center.

[0013] FIG. 5D depicts parallax differences for two adjacent cameras, relative to a center of perspective.

[0014] FIG. 5E depicts front color at an edge of an outer compressor lens element.

[0015] FIG. 6 depicts distortion correction curves plotted on a graph showing a percentage of distortion relative to a fractional field.

[0016] FIG. 7 depicts fields of view for adjacent cameras, including both Core and Extended fields of view (FOV), both of which can be useful for the design of an optimized panoramic multi-camera capture device.

[0017] FIG. 8A depicts a cross-sectional view of an improved imaging lens system having reduced parallax.

[0018] FIG. 8B depicts a graph of the calculated residual for a center of a perspective variation for the lens of FIG. 8A.

[0019] FIG. 8C-1 and FIG. 8C-2 provide lens prescription data for the lens of FIG. 8A.

[0020] FIG. 8D provides the calculated resolution, as a graph of MTF, for the lens of FIG. 8A.

[0021] FIG. 8E provides the calculated astigmatism and distortion at the image plane, as graphs, for the lens of FIG. 8A.

[0022] FIG. 8F provides a graph of the calculated relative illumination for the lens of FIG. 8A.

[0023] FIG. 8G provides an illustration of the front color for the lens of FIG. 8A.

[0024] FIG. 9A provides cross-sectional views of an alternate low parallax lens design.

[0025] FIG. 9B depicts a cross-sectional view of an expanded portion of FIG. 9A.

[0026] FIG. 10A provides cross-sectional views of a second alternate low parallax lens design.

[0027] FIG. 10B provides an alternate cross-sectional of the lens of FIG. 10A.

[0028] FIG. 10C provides the calculated resolution, as a graph of MTF, for the lens of FIG. 10A.

[0029] FIG. 10D provides a graph of the calculated relative illumination for the lens of FIG. 10A.

[0030] FIG. 11 provides cross-sectional views of a third alternate low parallax lens design.

[0031] FIG. 12 provides cross-sectional views of a fourth alternate low parallax lens design.

[0032] FIG. 13 depicts a cross-sectional view of an improved opto-mechanics construction for a multi-camera capture device, and a 3D view of a camera channel thereof.

[0033] FIG. 14 depicts a cross-sectional view of a seam and two adjacent stepped edge angle compressor lenses, near a seam, and a zoomed in view of a portion (B) thereof.

[0034] FIG. 15 depicts an electronics system diagram for a multi-camera capture device.

[0035] FIG. 16 depicts different views of an outer compressor lens element with lens datums, and an associated lens housing with datum features.

[0036] FIG. 17 depicts a cross-sectional view of a portion of an assembly of an improved multi-camera panoramic image capture device, and a portion thereof in detail.

[0037] FIG. 18A depicts a series of views of components and features of a lens housing and a channel loading support, that can be used during assembly of an improved multi-camera panoramic image capture device.

[0038] FIG. 18B-1 depicts a cross-sectional views of an alternate version of the lens housings and the interaction thereof, at or near a seam, to that shown in FIG. 14.

[0039] FIG. 18B-2 depicts a cross-sectional views of an alternate version of the channel to channel datums to those shown in FIG. 18A.

[0040] FIG. 19 depicts a view of another alternate design for the mounting of the camera channels to a central support.

[0041] FIG. 20 depicts a cross-sectional view of an optical system in which a low parallax imaging lens is combined with a relay optical system.

DETAILED DESCRIPTION

[0042] As is generally understood in the field of optics, a lens or lens assembly typically comprises a system or device having multiple lens elements which are mounted into a lens barrel or housing, and which work together to produce an optical image. An imaging lens captures a portion of the light coming from an object or plurality of objects that reside in object space at some distance(s) from the lens system. The imaging lens can then form an image of these objects at an output “plane”; the image having a finite size that depends on the magnification, as determined by the focal length of the imaging lens and the conjugate distances to the object(s) and image plane, relative to that focal length. The amount of image light that transits the lens, from object to image, depends in large part on the size of the aperture stop of the imaging lens, which is typically quantified by one or more values for a numerical aperture (NA) or an f-number (F# or F/#).

[0043] The image quality provided by the imaging lens depends on numerous properties of the lens design, including the selection of optical materials used in the design, the size, shapes (or curvatures) and thicknesses of the lens elements, the relative spacing of the lens elements one to another, the spectral bandwidth, polarization, light load (power or flux) of the transiting light, optical diffraction or scattering, and/or lens manufacturing tolerances or errors. The image quality is typically described or quantified in terms of lens aberrations (e.g., spherical, coma, or distortion), or the relative size of the resolvable spots provided by the lens, which is also often quantified by a modulation transfer function (MTF).

[0044] In a typical electronic or digital camera, an image sensor is nominally located at the image plane. This image sensor is typically a CCD or CMOS device, which is physically attached to a

heat sink or other heat removal means, and also includes electronics that provide power to the sensor, and read-out and communications circuitry that provide the image data to data storage or image processing electronics. The image sensor typically has a color filter array (CFA), such as a Bayer filter within the device, with the color filter pixels aligned in registration with the image pixels to provide an array of RGB (Red, Green, Blue) pixels. Alternative filter array patterns, including the CYGM filter (cyan, yellow, green, magenta) or an RGBW filter array (W=white), can be used instead. The image sensor typically includes a cover glass that can seal or protect the sensor from the environment. The cover glass can also provide UV or IR cut filtering by means of a thin film interference or dichroic coating, or that function can be provided on a separate window, external filter. The UV or IR cut filter reduces the level of the non-visible light incident, that is accompanying the image light to the sensor. Alternately, or in additionally, UV and IR cut filtering can be provided with a coating applied to a lens element, including to the outer surface of an outer lens element. The cover glass or filter can also be a UV light absorbing glass and provide UV filtering by a combination of absorption and coating reflectance.

[0045] It is noted that for many image sensors, the image light is incident directly onto the sensor pixels. However, many commercially available image sensors are further equipped with an integrated lenslet array, overlaying, and aligned to, the pixel array. A given lenslet directs its portion of incident image light to the corresponding underlying pixel. This approach can reduce optical crosstalk between pixels, as the incident image light is more likely to be photo-electrically converted into an electrical signal within the incident pixel than may have been true otherwise. As the pixels have become progressively smaller, this approach has become increasingly common, as a means to improve or maintain image resolution (MTF). Such sensors with integrated lenslet arrays can be used within the improved low parallax cameras (320) of the present approach.

[0046] In typical use, many digital cameras are used by people or remote systems in relative isolation, to capture images or pictures of a scene, without any dependence or interaction with any other camera devices. In some cases, such as surveillance or security, the operation of a camera may be directed by people or algorithms based on image content seen from another camera that has already captured overlapping, adjacent, or proximate image content. In another example, people capture panoramic images of a scene with an extended or wide FOV, such as a landscape scene, by sequentially capturing a sequence of adjacent images, while manually or automatically moving or pivoting to frame the adjacent images. Afterwards, image processing software, such as Photoshop or Lightroom, can be used to stitch, mosaic, or tile the adjacent images together to portray the larger extended scene. Image stitching or photo stitching is the process of combining multiple photographic images with overlapping fields of view to produce a segmented panorama or high-resolution image. Image quality improvements, including exposure or color corrections, can also be applied, either in real time, or in a post processing or image rendering phase, or a combination thereof.

[0047] Unless the objects in a scene are directionally illuminated and/or have a directional optical response (e.g., such as with reflectance), the available light is plenoptic, meaning that there is light ($\square$), travelling in every direction, or nearly so, in a given space or environment. A camera can then sample a subset of this light, as image light, with which it provides a resulting image that shows a given view or perspective of the different objects in the scene at one or more instants in time. If the camera is moved to a different nearby location and used to capture another image of part of that same scene, both the apparent perspectives and relative positioning of the objects will change. In the latter case, one object may now partially occlude another, while a previously hidden object becomes at least partially visible. These differences in the apparent position or direction of an object are known as parallax. In particular, parallax is a displacement or difference in the apparent position of an object viewed along two different lines of sight and is measured by the angle or semi-angle of inclination between those two lines.

[0048] In a stereoscopic image capture or projection system, dual view parallax is a cue, along with

shadowing, occlusion, and perspective, that can provide a sense of depth. For example, in a stereo (3D) projection system, polarization or spectrally encoded image pairs can be overlap projected onto a screen to be viewed by audience members wearing appropriate glasses. The amount of parallax can have an optimal range, outside of which, the resulting sense of depth can be too small to really be noticed by the audience members, or too large to properly be fused by the human visual system.

[0049] Whereas, in a panoramic image capture application, parallax differences can be regarded as an error that can complicate both image stitching and appearance. In the example of an individual manually capturing a panoramic sequence of landscape images, the visual differences in perspective or parallax across images may be too small to notice if the objects in the scene are sufficiently distant (e.g., optically at infinity). An integrated panoramic capture device with a rotating camera or multiple cameras has the potential to continuously capture real time image data at high resolution without being dependent on the uncertainties of manual capture. But such a device can also introduce its own visual disparities, image artifacts, or errors, including those of parallax, perspective, and exposure. Although the resulting images can often be successfully stitched together with image processing algorithms, the input image errors complicate and lengthen image processing time, while sometimes leaving visually obvious residual errors.

[0050] To provide context, FIG. 1 depicts a portion of an integrated panoramic multi-camera capture device **100** having two adjacent cameras **120** in housings **130** which are designed for reduced parallax image capture. These cameras are alternately referred to as image pick-up units, or camera channels, or objective lens systems. The cameras **120** each have a plurality of lens elements (see FIG. 2) that are mounted within a lens barrel or housing **130**. The adjacent outer lens elements **137** have adjacent beveled edges **132** and are proximately located, one camera channel to another, but which may not be in contact, and thus are separated by a gap or seam **160** of finite width. Some portion of the available light ($\square$), or light rays **110**, from a scene or object space **105** will enter a camera **120** to become image light that was captured within a constrained FOV and directed to an image plane, while other light rays will miss the cameras entirely. Some light rays **110** will propagate into the camera and transit the constituent lens elements as edge-of-field chief rays **170**, or perimeter rays, while other light rays can potentially propagate through the lens elements to create stray or ghost light and erroneous bright spots or images. As an example, some light rays (**167**) that are incident at large angles to the outer surface of an outer lens element **137** can transit a complex path through the lens elements of a camera and create a detectable ghost image at the image plane **150**.

[0051] In greater detail, FIG. 2A depicts a cross-section of part of a camera **120** having a set of lens elements **135** mounted in a housing (**130**, not shown) within a portion of an integrated panoramic multi-camera capture device **100**. A fan of light rays **110** from object space **105**, spanning the range from on axis to full field off axis chief rays, are incident onto the outer lens element **137**, and are refracted and transmitted inwards. This image light **115** that is refracted and transmitted through further inner lens elements **140**, through an aperture stop **145**, converges to a focused image at or near an image plane **150**, where an image sensor (not shown) is typically located. The lens system **120** of FIG. 2A can also be defined as having a lens form that consists of outer lens element **137** or compressor lens element, and inner lens elements **140**, the latter of which can also be defined as consisting of a pre-stop wide angle lens group, and a post-stop eyepiece-like lens group. This compressor lens element (**137**) directs the image light **115** sharply inwards, compressing the light, to both help enable the overall lens assembly to provide a short focal length, while also enabling the needed room for the camera lens housing or barrel to provide the mechanical features necessary to both hold or mount the lens elements and to interface properly with the barrel or housing of an adjacent camera. The image light that transited a camera lens assembly from the outer lens element **137** to the image plane **150** will provide an image having an image quality, that can be quantified by an image resolution, image contrast, a depth of focus, and other attributes, whose quality was

defined by the optical aberrations (e.g., astigmatism, distortion, or spherical) and chromatic or spectral aberrations, encountered by the transiting light at each of the lens elements (**137**, **140**) within a camera **120**. FIG. 2B depicts a fan of chief rays **170**, or perimeter rays, incident along or near a beveled edge **132** of the outer lens element **137** of the camera optics (**120**) depicted in FIG. 2A. FIG. 2B also depicts a portion of a captured, polygonal shaped or asymmetrical, FOV **125**, that extends from the optical axis **185** to a line coincident with an edge ray.

[0052] In the camera lens design depicted in FIG. 2A, the outer lens element **137** functions as a compressor lens element that redirects the transiting image light **115** towards a second lens element **142**, which is the first lens element of the group of inner lens elements **140**. In this design, this second lens element **142** has a very concave shape that is reminiscent of the outer lens element used in a fish-eye type imaging lens. This compressor lens element directs the image light **115** sharply inwards, or bends the light rays, to both help enable the overall lens assembly to provide a short focal length, while also enabling the needed room for the camera lens housing **130** or barrel to provide the mechanical features necessary to both hold or mount the lens elements **135** and to interface properly with the barrel or housing of an adjacent camera. However, with a good lens and opto-mechanical design, and an appropriate sensor choice, a camera **120** can be designed with a lens assembly that supports an image resolution of 20-30 pixels/degree, to as much as 110 pixels/degree, or greater, depending on the application and the device configuration.

[0053] The resultant image quality from these cameras will also depend on the light that scatters at surfaces, or within the lens elements, and on the light that is reflected or transmitted at each lens surface. The surface transmittance and camera lens system efficiency can be improved by the use of anti-reflection (AR) coatings. The image quality can also depend on the outcomes of non-image light. Considering again FIG. 1, other portions of the available light can be predominately reflected off of the outer lens element **137**. Yet other light that enters a camera **120** can be blocked or absorbed by some combination of blackened areas (not shown) that are provided at or near the aperture stop, the inner lens barrel surfaces, the lens element edges, internal baffles or light trapping features, a field stop, or other surfaces. Yet other light that enters a camera can become stray light or ghost light **167** that is also potentially visible at the image plane.

[0054] The aggregate image quality obtained by a plurality of adjacent cameras **120** within an improved integrated panoramic multi-camera capture device **100** (e.g., FIG. 1) can also depend upon a variety of other factors including the camera to camera variations in the focal length and/or track length, and magnification, provided by the individual cameras. These parameters can vary depending on factors including the variations of the glass refractive indices, variations in lens element thicknesses and curvatures, and variations in lens element mounting. As an example, images that are tiled or mosaiced together from a plurality of adjacent cameras will typically need to be corrected, one to the other, to compensate for image size variations that originate with camera magnification differences (e.g., $\pm 2\%$).

[0055] The images produced by a plurality of cameras in an integrated panoramic multi-camera capture device **100** can also vary in other ways that effect image quality and image mosaicing or tiling. In particular, the directional pointing or collection of image light through the lens elements to the image sensor of any given camera **120** can vary, such that the camera captures an angularly skewed or asymmetrical FOV (FOV.Math.) or mis-sized FOV (FOV $\pm$). The lens pointing variations can occur during fabrication of the camera (e.g., lens elements, sensor, and housing) or during the combined assembly of the multiple cameras into an integrated panoramic multi-camera capture device **100**, such that the alignment of the individual cameras is skewed by misalignments or mounting stresses. When these camera pointing errors are combined with the presence of the seams **160** between cameras **120**, images for portions of an available landscape or panoramic FOV that may be captured, may instead be missed or captured improperly. The variabilities of the camera pointing, and seams can be exacerbated by mechanical shifts and distortions that are caused by internal or external environmental factors, such as heat or light (e.g., image content), and

particularly asymmetrical loads thereof.

[0056] In comparison to the FIG. 1 system, in a typical commercially available panoramic camera, the seams between cameras are outright gaps that can be 30-50 mm wide, or more. In particular, as shown in FIG. 3, a panoramic multi-camera capture device **101** can have adjacent cameras **120** or camera channels separated by large gaps or seams **160**, between which there are blind spots or regions **165** from which neither camera can capture images. The actual physical seams **160** between adjacent camera channels or outer lens elements **137** (FIG. 1 and FIG. 3) can be measured in various ways; as an actual physical distance between adjacent lens elements or lens housings, as an angular extent of lost FOV, or as a number of “lost” pixels. However, the optical seam, as the distance between outer chief rays of one camera to another can be larger yet, due to any gaps in light acceptance caused by vignetting or coating limits. For example, anti-reflection (AR) coatings are not typically deposited to the edges of optics, but an offsetting margin is provided, to provide a coated clear aperture (CA).

[0057] To compensate for both camera misalignments and the large seams **160**, and to reduce the size of the blind regions **165**, the typical panoramic multi-camera capture devices **101** (FIG. 3) have each of the individual cameras **120** capture image light **115** from wide FOVs **125** that provide overlap **127**, so that blind regions **165** are reduced, and the potential capturable image content that is lost is small. As another example, in most of the commercially available multi-camera capture devices **101**, the gaps are 25-50+mm wide, and the compensating FOV overlap between cameras is likewise large; e.g., the portions of the FOVs **125** that are overlapping and are captured by two adjacent cameras **120** can be as much as 10-50% of a camera's FOV. The presence of such large image overlaps from shared FOVs **125** wastes potential image resolution and increases the image processing and image stitching time, while introducing significant image parallax and perspective errors. These errors complicate image stitching, as the errors must be corrected or averaged during the stitching process. In such systems, the parallax is not predictable because it changes as a function of object distance. If the object distance is known, the parallax can be predicted for given fields of view and spacing between cameras. But because the object distance is not typically known, parallax errors then complicate image stitching. Optical flow and common stitching algorithms determine an object depth and enable image stitching, but with processing power and time burdens.

[0058] Similarly, in a panoramic multi-camera capture device **100**, of the type of FIG. 1, with closely integrated cameras, the width and construction at the seams **160** can be an important factor in the operation of the entire device. However, the seams can be made smaller than in FIG. 3, with the effective optical seam width between the FOV edges of two adjacent cameras determined by both optical and mechanical contributions. For example, by using standard optical engineering practices to build lens assemblies in housings, the mechanical width of the seams **160** between the outer lens elements **137** of adjacent cameras might be reduced to 4-6 mm. For example, it is standard practice to assemble lens elements into a lens barrel or housing that has a minimum radial width of 1-1.5 mm, particularly near the outermost lens element. Then accounting for standard coated clear apertures or coating margins, and accounting for possible vignetting, aberrations of the entrance pupil, front color, chip edges, and trying to mount adjacent lens assemblies or housings in proximity by standard techniques. Thus, when accounting for both optics and mechanics, an optical seam width between adjacent lenses can easily be 8-12 mm or more.

[0059] But improved versions of the panoramic multi-camera capture device (**300**) of the type of FIG. 1, with optical and opto-mechanical designs that enable significantly smaller seams, and with further improved parallax performance, are possible. As a first example, for the present technology for improved polygonal shaped cameras, during early stages of fabrication of outer lens elements **137**, these lenses can have a circular shape and can be AR coated to at or near their physical edges. When these lenses are subsequently processed to add the polygonal shape defining beveled edges **132** (e.g., FIG. 2B), a result can be that the AR coatings will essentially extend to the beveled lens

edges. The effective optical or coated edge apertures can then be defined by any allowances for mechanical mounting or for the standard edge grind that is used in optics manufacturing to avoid edge chipping. With this approach, and a mix of other techniques that will be subsequently discussed, the optical seams can be reduced to 1-5 mm width.

[0060] The use of an improved multi-camera capture device **300** having improved low-parallax cameras **320** is shown in context in FIG. **15**, which in part depicts an electronics system diagram. In this example, a dodecahedron type device has **11** cameras **320**, and an electro-mechanical interface in the twelfth camera position. Image data can be collected from each of the **11** cameras, and directed through an interface input-output module, through a cable or bundle of cables, to a portable computer. A primary goal of the present invention is to produce high quality low-parallax panoramic images from an improved multi-camera panoramic capture device (**300**) used in a system such as that of FIG. **15**. This broad goal can be enabled by developing a systemic range of design strategies to inform both the optical and opto-mechanical lens design efforts, and the opto-mechanical device design and fabrication efforts, as well as strategies for improved image capture and processing. This goal can also be enabled by providing for both initial and ongoing camera and device calibration. In broad terms, the image processing or rendering of images is a method to generate quality images from the raw captured image data that depends on the camera intrinsics (geometric factors such as focal length and distortion), the camera extrinsics (geometric factors such as camera orientation to object space), other camera parameters such as vignetting and transmission, and illumination parameters such as color and directionality. With respect to an improved multi-camera panoramic capture device **300**, knowledge of the camera intrinsics for any given camera **320** in a device, all assists towards completing reliable and repeatable tiling of images obtained from a plurality of adjacent cameras. Thus the subsequent discussions are broadly focused on providing optical (camera or objective lens) designs that can enable the desired image quality, as well as camera and device assembly approaches, management of key tolerances, camera calibration, knowledge of camera intrinsics and extrinsics, and other factors that can likewise affect the resultant device performance. The improved panoramic multi-camera capture devices of the present invention can be used to support a wide variety of applications or markets, including cinematic image capture, augmented reality or virtual reality (VR) image capture, surveillance or security imaging, sports or event imaging, mapping or photogrammetry, vehicular navigation, and robotics.

[0061] Before exploring opto-mechanical means for enabling improved panoramic multi-camera capture devices (**300**), means for providing cameras **120** that are improved for use in these systems are developed. Accordingly, the goals include providing improved cameras (**320**, **920**) having both reduced parallax errors and image overlap. As one aspect of the present approach, a goal is to reduce the residual parallax error for the edge chief rays collected respectively by each camera in an adjacent pair. The parallax error is defined as the change in parallax with respect to object distance (e.g., that the chief ray trajectory with respect to a near distance (e.g., 3 feet) from the device, versus a far distance (e.g., 1 mile), is slightly different). For example, as one goal or target for reduced parallax, or to have effectively no parallax error, or to be “parallax-free”, is that the chief rays of adjacent cameras should deviate from parallelism to each other by ≤ 0.5 - 2.0 deg., and preferably by ≤ 0.01 - 0.1 deg. Alternately, or equivalently, the parallax error, as assessed as a perspective error in terms of location on the image plane, should be reduced to ≤ 2 pixels, and preferably to ≤ 0.5 pixel. As another aspect of the present approach, the width of the seams **160** between adjacent cameras (e.g., **120**, **320**, **920**) assembled into their own lens housings are to be reduced. The goal is to reduce the width of the seams, both in terms of their absolute physical width, and their optical width or an effective width. For example, a goal is to reduce a seam **160** between adjacent outer lens elements **137** to having a maximum gap or an actual physical seam width in a range of only ≈ 0.5 - 3.0 mm, and to then reduce the maximum optical seam width to a range of about only 1-6 mm. As an example, these reduced seams widths can translate to a reduced

angular extent of lost FOV of only 0.25-1.0°, or a number of “lost” pixels of only 2-20 pixels. For example, for a device providing 8k pixels around a 360-degree panorama equirectangular image, a loss of only 2-4 pixels at the seams can be acceptable as the residual image artifacts can be difficult to perceive. The actual details or numerical targets for effectively no-parallax error, or for the maximum optical seam width, depend on many factors including the detailed opto-mechanical designs of the improved cameras (**320**) and overall device (**300**), management of tolerances, possible allowances for a center offset distance or an amount of extended FOV (**215**) and the targets for low parallax therein, and the overall device specifications (e.g., diameter, sensor resolution or used sensor pixels within an imaged FOV or a Core FOV **205** (FIG. 7)). Further goals, enabled by some combination of the above improvements, are for each camera to reliably and quickly provide output images from an embedded sensor package that are cropped down to provide core FOV images, and then that each cropped image can be readily seamed or tiled with cropped images provided by adjacent cameras, so as to readily provide panoramic output images from an improved multi-camera capture device (**300**) in real time.

[0062] An improved panoramic multi-camera capture device **300**, such as that of FIG. **13** and FIG. **15**, can have a plurality of cameras arranged around a circumference of a sphere to capture a 360-degree annular FOV. Alternately, a panoramic multi-camera capture device can have a plurality of cameras arranged around a spherical or polyhedral shape. A polyhedron is a three-dimensional solid consisting of a collection of polygons that are contiguous at the edges. One polyhedral shape, as shown in FIG. **4**, is that of a dodecahedron **50**, which has 12 sides or faces, each shaped as a regular pentagon **55**, and 20 vertices or corners (e.g., a vertex **60**). A panoramic multi-camera capture device formed to the dodecahedron shape has cameras with a pentagonally shaped outer lens elements that nominally image a 69.1° full width field of view. Another shape is that of a truncated icosahedron, like a soccer ball, which as is also shown in FIG. **4**, and has a combination of 12 regular pentagonal sides or faces, 20 regular hexagonal sides or faces, 60 vertices, and 90 edges. More complex shapes, with many more sides, such as regular polyhedra, Goldberg polyhedra, or shapes with octagonal sides, or even some irregular polyhedral shapes, can also be useful. For example, a Goldberg chamfered dodecahedron is similar to the truncated icosahedron, with both pentagonal and hexagonal facets, totaling 42 sides. But in general, the preferred polyhedrons for the current purpose have sides or faces that are hexagonal or pentagonal, which are generally roundish shapes with beveled edges **132** meeting at obtuse corners. Other polyhedral shapes, such as an octahedron or a regular icosahedron can be used, although they have triangular facets. Polyhedral facets with more abrupt or acute corners, such as square or triangular faces, can be easier to fabricate, as compared to facets with pentagonal and or hexagonal facets, as they have fewer edges to cut to provide polygonal edges on the outermost lens element, so as to define a captured polygonal FOV. However, greater care can then be needed in cutting, beveling, and handling the optic because of those acute corners. Additionally, for lens facets with large FOVs and acute facet angles, it can be more difficult to design the camera lenses and camera lens housings for optical and opto-mechanical performance. Typically, a 360° polyhedral camera will not capture a full spherical FOV as at least part of one facet is sacrificed to allow for support features and power and communications cabling, such as via a mounting post. However, if the device communicates wirelessly, and is also hung by a thin cable to a vertex, the FOV lost to such physical connections can be reduced.

[0063] As depicted in FIG. **1** and FIG. **2B**, a camera channel **120** can resembles a frustum, or a portion thereof, where a frustum is a geometric solid (normally a cone or pyramid) that lies between one or two parallel planes that cut through it. In that context, a fan of chief rays **170** corresponding to a polygonal edge, can be refracted by an outer compressor lens element **137** to nominally match the frustum edges in polyhedral geometries.

[0064] To help illustrate some issues relating to camera geometry, FIG. **5A** illustrates a cross-sections of a pentagonal lens **175** capturing a pentagonal FOV **177** and a hexagonal lens **180**

capturing a hexagonal FOV **182**, representing a pair of adjacent cameras whose outer lens elements have pentagonal and hexagonal shapes, as can occur with a truncated icosahedron, or soccer ball type panoramic multi-camera capture devices (e.g., **100**, **300**). The theoretical hexagonal FOV **182** spans a half FOV of 20.9° , or a full FOV of 41.8° (□.sub.1) along the sides, although the FOV near the vertices is larger. The pentagonal FOV **177** supports 36.55° FOV (□.sub.2) within a circular region, and larger FOVs near the corners or vertices. Notably, in this cross-section, the pentagonal FOV **177** is asymmetrical, supporting a 20-degree FOV on one side of an optical axis **185**, and only a 16.5-degree FOV on the other side of the optical axis.

[0065] Optical lenses are typically designed using programs such as ZEMAX or Code V. Design success typically depends, in part, on selecting the best or most appropriate lens parameters, identified as operands, to use in the merit function. This is also true when designing a lens system for an improved low-parallax multi-camera panoramic capture device (**300**), for which there are several factors that affect performance (including, particularly parallax) and several parameters that can be individually or collectively optimized, so as to control it. One approach targets optimization of the “NP” point, or more significantly, variants thereof.

[0066] As background, in the field of optics, there is a concept of the entrance pupil, which is a projected image of the aperture stop as seen from object space, or a virtual aperture which the imaged light rays from object space appear to propagate towards before any refraction by the first lens element. By standard practice, the location of the entrance pupil can be found by identifying a paraxial chief ray from object space **105**, that transits through the center of the aperture stop, and projecting or extending its object space direction forward to the location where it hits the optical axis **185**. In optics, incident Gauss or paraxial rays are understood to reside within an angular range $\leq 10^\circ$ from the optical axis, and correspond to rays that are directed towards the center of the aperture stop, and which also define the entrance pupil position. Depending on the lens properties, the entrance pupil may be bigger or smaller than the aperture stop, and located in front of, or behind, the aperture stop.

[0067] By comparison, in the field of low-parallax cameras, there is a concept of a no-parallax (NP) point, or viewpoint center. Conceptually, the “NP Point” has been associated with a high FOV chief ray or principal ray incident at or near the outer edge of the outermost lens element, and projecting or extending its object space direction forward to the location where it hits the optical axis **185**. For example, depending on the design, camera channels in a panoramic multi-camera capture device can support half FOVs with non-paraxial chief rays at angles $>31^\circ$ for a dodecahedron type system (FIG. 4) or $>20^\circ$ for a truncated icosahedron type system (see FIG. 4 and FIG. 5A). This concept of the NP point projection has been applied to the design of panoramic multi-camera capture devices, relative to the expectations for chief ray propagation and parallax control for adjacent optical systems (cameras). It is also stated that if a camera is pivoted about the NP point, or a plurality of camera's appear to rotate about a common NP point, then parallax errors will be reduced, and images can be aligned with little or no parallax error or perspective differences. But in the field of low parallax cameras, the NP point has also been equated to the entrance pupil, and the axial location of the entrance pupil that is estimated using a first order optics tangent relationship between a projection of a paraxial field angle and the incident ray height at the first lens element (see FIGS. 2A, 2B).

[0068] Thus, confusingly, in the field of designing of low-parallax cameras, the NP point has also been previously associated with both with the projection of edge of FOV chief rays and the projection of chief rays that are within the Gauss or paraxial regime. As will be seen, in actuality, they both have value. In particular, an NP point associated with the paraxial entrance pupil can be helpful in developing initial specifications for designing the lens, and for describing the lens. An NP point associated with non-paraxial edge of field chief rays can be useful in targeting and understanding parallax performance and in defining the conical volume or frustum that the lens assembly can reside in.

[0069] The projection of these non-paraxial chief rays can miss the paraxial chief ray defined entrance pupil because of both lens aberrations and practical geometry related factors associated with these lens systems. Relative to the former, in a well-designed lens, image quality at an image plane is typically prioritized by limiting the impact of aberrations on resolution, telecentricity, and other attributes. Within a lens system, aberrations at interim surfaces, including the aperture stop, can vary widely, as the emphasis is on the net sums at the image plane. Aberrations at the aperture stop are often somewhat controlled to avoid vignetting, but a non-paraxial chief ray need not transit the center of the aperture stop or the projected paraxially located entrance pupil.

[0070] To expand on these concepts, and to enable the design of improved low parallax lens systems, it is noted that the camera lens system **120** in FIG. 2A depicts both a first NP point **190A**, corresponding to the entrance pupil as defined by a vectoral projection of paraxial chief rays from object space **105**, and an offset second NP point **190B**, corresponding to a vectoral projection of a non-paraxial chief rays from object space. Both of these ray projections cross the optical axis **185** in locations behind both the lens system and the image plane **150**. As will be subsequently discussed, the ray behavior in the region between and proximate to the projected points **190A** and **190B** can be complicated and neither projected location or point has a definitive value or size. A projection of a chief ray will cross the optical axis at a point, but a projection of a group of chief rays will converge towards the optical axis and cross at different locations, that can be tightly clustered (e.g., within a few or tens of microns), where the extent or size of that “point” can depends on the collection of proximate chief rays used in the analysis. Whereas, when designing low parallax imaging lenses that image large FOVs, the axial distance or difference between the NP points **190A** and **190B** that are provided by the projected paraxial and non-paraxial chief rays can be significantly larger (e.g., millimeters). Thus, as will also be discussed, the axial difference represents a valuable measure of the parallax optimization (e.g., a low parallax volume **188**) of a lens system designed for the current panoramic capture devices and applications. As will also be seen, the design of an improved device (**300**) can be optimized to position the geometric center of the device, or device center **196**, outside, but proximate to this low parallax volume **188**, or alternately within it, and preferably proximate to a non-paraxial chief ray NP point.

[0071] As one aspect, FIG. 5A depicts the projection of the theoretical edge of the fields of view (FOV edges **155**), past the outer lens elements (lenses **175** and **180**) of two adjacent cameras, to provide lines directed to a common point (**190**). These lines represent theoretical limits of the complex “conical” opto-mechanical lens assemblies, which typically are pentagonally conical or hexagonally conical limiting volumes. Again, ideally, in a no-parallax multi-camera system, the entrance pupils or NP points of two adjacent cameras are co-located. But to avoid mechanical conflicts, the mechanics of a given lens assembly, including the sensor package, should generally not protrude outside a frustum of a camera system and into the conical space of an adjacent lens assembly. However, real lens assemblies in a multi-camera panoramic capture device are also separated by seams **160**. Thus, the real chief rays **170** that are accepted at the lens edges, which are inside of both the mechanical seams and a physical width or clear aperture of a mounted outer lens element (lenses **175** and **180**), when projected generally towards a paraxial NP point **190**, can land instead at offset NP points **192**, and be separated by an NP point offset distance **194**.

[0072] This can be better understood by considering the expanded area A-A in proximity to a nominal or ideal point NP **190**, as shown in detail in FIG. 5B. Within a hexagonal FOV **182**, light rays that propagate within the Gauss or paraxial region (e.g., paraxial ray **173**), and that pass through the nominal center of the aperture stop, can be projected to a nominal NP point **190** (corresponding to the entrance pupil), or to an offset NP point **190A** at a small NP point difference or offset **193** from a nominal NP point **190**. Whereas, the real hexagonal lens edge chief rays **170** associated with a maximum inscribed circle within a hexagon, can project to land at a common offset NP point **192A** that can be at a larger offset distance (**194A**). The two adjacent cameras in FIGS. 5A,B also may or may not share coincident NP points (e.g., **190**). Distance offsets can occur

due to various reasons, including geometrical concerns between cameras (adjacent hexagonal and pentagonal cameras), geometrical asymmetries within a camera (e.g., for a pentagonal camera), or from limitations from the practical widths of seams **160**, or because of the directionality difference amongst aberrated rays.

[0073] As just noted, there are also potential geometric differences in the projection of incident chief rays towards a simplistic nominal “NP point” (**190**). First, incident imaging light paths from near the corners or vertices or mid-edges (mid-chords) of the hexagonal or pentagonal lenses may or may not project to common NP points within the described range between the nominal paraxial NP point **190** and an offset NP point **192B**. Also, as shown in FIG. 5B, just from the geometric asymmetry of the pentagonal lenses, the associated pair of edge chief rays **170** and **171** for the real accepted FOV, can project to different nominal NP points **192B** that can be separated from both a paraxial NP point (**190**) by an offset distance **194B** and from each other by an offset distance **194C**.

[0074] As another issue, during lens design, the best performance typically occurs on axis, or near on axis (e.g., ≤ 0.3 field (normalized)), near the optical axis **185**. In many lenses, good imaging performance, by design, often occurs at or near the field edges, where optimization weighting is often used to force compliance. The worst imaging performance can then occur at intermediate fields (e.g., 0.7-0.8 of a normalized image field height). Considering again FIG. 5A,B, intermediate off axis rays, from intermediate fields ($\square$) outside the paraxial region, but not as extreme as the edge chief rays ($10^\circ < \square < 20.9^\circ$), can project towards intermediate NP points between a nominal NP point **190** and an offset NP point **192B**. But other, more extreme off axis rays, particularly from the 0.7-0.8 intermediate fields, that are more affected by aberrations, can project to NP points at locations that are more or less offset from the nominal NP point **190** than are the edge of field offset NP points **192B**. Accounting for the variations in lens design, the non-paraxial offset “NP” points can fall either before (closer to the lens) the paraxial NP point (the entrance pupil) as suggested in FIG. 5B, or after it (as shown in FIG. 2A).

[0075] This is shown in greater detail in FIG. 5C, which essentially illustrates a further zoomed-in region A-A of FIG. 5B, but which illustrates an impact from vectoral projected ray paths associated with aberrated image rays, that converge at and near the paraxial entrance pupil (**190**), for an imaging lens system that was designed and optimized using the methods of the present approach. In FIG. 5C, the projected ray paths of green aberrated image rays at multiple fields from a camera lens system converge within a low parallax volume **188** near one or more “NP” points. Similar illustrations of ray fans can also be generated for Red or Blue light. The projection of paraxial rays **173** can converge at or near a nominal paraxial NP point **190**, or entrance pupil, located on a nominal optical axis **185** at a distance Z behind the image plane **150**. The projection of edge of field rays **172**, including chief rays **171**, converge at or near an offset NP point **192B** along the optical axis **185**. The NP point **192B** can be quantitatively defined, for example, as the center of mass of all edge of field rays **172**. An alternate offset NP point **192A** can be identified, that corresponds to a “circle of least confusion”, where the paraxial, edge, and intermediate or mid-field rays, aggregate to the smallest spot. These different “NP” points are separated from the paraxial NP point by offset distances **194A** and **194B**, and from each other by an offset distance **194C**. Thus, it can be understood that an aggregate “NP point” for any given real imaging lens assembly or camera lens that supports a larger than paraxial FOV, or an asymmetrical FOV, is typically not a point, but instead can be an offset low parallax (LP) smudge or volume **188**.

[0076] Within a smudge or low parallax volume **188**, a variety of possible optimal or preferred NP points can be identified. For example, an offset NP point corresponding to the edge of field rays **172** can be emphasized, so as to help provide improved image tiling. An alternate mid-field (e.g., 0.6-0.8) NP point (not shown) can also be tracked and optimized for. Also the size and position of the overall “LP” smudge or volume **188**, or a preferred NP point (e.g., **192B**) therein, can change depending on the lens design optimization. Such parameters can also vary amongst lenses, for one fabricated lens system of a given design to another, due to manufacturing differences amongst lens

assemblies. Although FIG. 5C depicts these alternate offset “NP points” **192A,B** for non-paraxial rays as being located after the paraxial NP point **190**, or further away from the lens and image plane, other lenses of this type, optimized using the methods of the present approach, can be provided where similar non-paraxial NP points **192A,B** that are located with a low parallax volume **188** can occur at positions between the image plane and the paraxial NP point.

[0077] FIG. 5C also shows a location for a center of the low-parallax multi-camera panoramic capture device, device center **196**. Based on optical considerations, an improved panoramic multi-camera capture device **300** can be preferably optimized to nominally position the device center **196** within the low parallax volume **188**. Optimized locations therein can include being located at or proximate either of the offset NP points **192A** or **192B**, or within the offset distance **194B** between them, so as to prioritize parallax control for the edge of field chief rays. The actual position therein depends on parallax optimization, which can be determined by the lens optimization relative to spherical aberration of the entrance pupil, or direct chief ray constraints, or distortion, or a combination thereof. For example, whether the spherical aberration is optimized to be over corrected or under corrected, and how weightings on the field operands in the merit function are used, can affect the positioning of non-paraxial “NP” points for peripheral fields or mid fields. The “NP” point positioning can also depend on the management of fabrication tolerances and the residual variations in lens system fabrication. The device center **196** can also be located proximate to, but offset from the low parallax volume **188**, by a center offset distance **198**. This approach can also help tolerance management and provide more space near the device center **196** for cables, circuitry, cooling hardware, and the associated structures. In such case, the adjacent cameras **120** can then have offset low parallax volumes **188** of “NP” points (FIG. 5D), instead of coincident ones (FIGS. 5A, B). In this example, if the device center **196** is instead located at or proximate to the paraxial entrance pupil, NP point **190**, then effectively one or more of the outer lens elements **137** of the cameras **120** are undersized and the desired full FOVs are not achievable. FIG. **10B** depicts the possible positioning of a similar lens system **920** with respect to an offset device center **910**.

[0078] Thus, while the no-parallax (NP) point is a useful concept to work towards, and which can valuably inform panoramic image capture and systems design, and aid the design of low-parallax error lenses, it is idealized, and its limitations must also be understood. Considering this discussion of the NP point(s) and LP smudges, in enabling an improved low-parallax multi-camera panoramic capture device (lens design examples to follow; e.g., device **920** of FIG. **8A**), it is important to understand ray behavior in this regime, and to define appropriate parameters or operands to optimize, and appropriate target levels of performance to aim for. In the latter case, for example, a low parallax lens with a track length of 65-70 mm can be designed for in which the LP smudge is as much as 10 mm wide (e.g., offset distance **194A**). But alternate lens designs, for which this parameter is further improved, can have a low parallax volume **188** with a longitudinal LP smudge width or width along the optical axis (offset **194A**) of a few millimeters or less.

[0079] The width and location of the low parallax volume **188**, and the vectoral directions of the projections of the various chief rays, and their NP point locations within a low parallax volume, can be controlled during lens optimization by a method using operands associated with a fan of chief rays **170** (e.g., FIGS. 2A,B). But the LP smudge or LP volume **188** of FIG. 5C can also be understood as being a visualization of the transverse component of spherical aberration of the entrance pupil, and this parameter can be used in an alternate, but equivalent, design optimization method to using chief ray fans. In particular, during lens optimization, using Code V for example, the lens designer can create a special user defined function or operand for the transverse component (e.g., ray height) of spherical aberration of the entrance pupil, which can then be used in a variety of ways. For example, an operand value can be calculated as a residual sum of squares (RSS) of values across the whole FOV or across a localized field, using either uniform or non-uniform weightings on the field operands. In the latter case of localized field preferences, the values can be

calculated for a location at or near the entrance pupil, or elsewhere within a low parallax volume **188**, depending on the preference towards paraxial, mid, or peripheral fields. An equivalent operand can be a width of a circle of least confusion in a plane, such as the plane of offset NP point **192A** or that of offset NP **192B**, as shown in FIG. 5C. The optimization operand can also be calculated with a weighting to reduce or limit parallax error non-uniformly across fields, with a disproportionate weighting favoring peripheral or edge fields over mid-fields. Alternately, the optimization operand can be calculated with a weighting to provide a nominally low parallax error in a nominally uniform manner across all fields (e.g., within or across a Core FOV **205**, as in FIG. 7). That type of optimization may be particularly useful for mapping applications.

[0080] Whether the low-parallax lens design and optimization method uses operands based on chief rays or spherical aberration of the entrance pupil, the resulting data can also be analyzed relative to changes in imaging perspective. In particular, parallax errors versus field and color can also be analyzed using calculations of the Center of Perspective (COP), which is a parameter that is more directly relatable to visible image artifacts than is a low parallax volume, and which can be evaluated in image pixel errors or differences for imaging objects at two different distances from a camera system. The center of perspective error is essentially the change in a chief ray trajectory given multiple object distances-such as for an object at a close distance (3 ft), versus another at “infinity”.

[0081] In drawings and architecture, perspective, is the art of drawing solid objects on a two-dimensional surface so as to give a correct impression of their height, width, depth, and position in relation to each other when viewed from a particular point. For example, for illustrations with linear or point perspective, objects appear smaller as their distance from the observer increases. Such illustrated objects are also subject to foreshortening, meaning that an object's dimensions along the line of sight appear shorter than its dimensions across the line of sight. Perspective works by representing the light that passes from a scene through an imaginary rectangle (realized as the plane of the illustration), to a viewer's eye, as if the viewer were looking through a window and painting what is seen directly onto the windowpane.

[0082] Perspective is related to both parallax and stereo perception. In a stereoscopic image capture or projection, with a pair of adjacent optical systems, perspective is a visual cue, along with dual view parallax, shadowing, and occlusion, that can provide a sense of depth. As noted previously, parallax is the visual perception that the position or direction of an object appears to be different when viewed from different positions. In the case of image capture by a pair of adjacent cameras with at least partially overlapping fields of view, parallax image differences are a cue for stereo image perception, or are an error for panoramic image assembly.

[0083] To capture images with an optical system, whether a camera or the human eye, the optical system geometry and performance impacts the utility of the resulting images for low parallax (panoramic) or high parallax (stereo) perception. In particular, for an ideal lens, all the chief rays from object space point exactly towards the center of the entrance pupil, and the entrance pupil is coincident with the center of perspective (COP) or viewpoint center for the resulting images. There are no errors in perspective or parallax for such an ideal lens.

[0084] But for a real lens, having both physical and image quality limitations, residual parallax errors can exist. As stated previously, for a real lens, a projection of the paraxial chief rays from the first lens element, will point towards a common point, the entrance pupil, and its location can be determined as an axial distance from the front surface of that first element. Whereas, for a real lens capturing a FOV large enough to include non-paraxial chief rays, the chief rays in object space can point towards a common location or volume near, but typically offset from, the center of the entrance pupil. These chief rays do not intrinsically coincide at a single point, but they can be directed through a small low parallax volume **188** (e.g., the LP “smudge”) by appropriate lens optimization. The longitudinal or axial variation of rays within the LP smudge can be determined from the position a chief ray crosses the optic axis. The ray errors can also be measured as a

transverse width or axial position of the chief rays within an LP smudge.

[0085] The concept of parallax correction, with respect to centers of perspective, is illustrated in FIG. 5D. A first camera lens **120A** collects and images light from object space **105** into at least a Core FOV, including light from two outer ray fans **179A** and **179B**, whose chief ray projections converge towards a low parallax volume **188A**. These ray fans can correspond to a group of near edge or edge of field rays **172**, as seen in FIG. 2B or FIG. 5C. As was shown in FIG. 5C, within an LP volume **188**, the vectorial projection of such rays from object space, generally towards image space, can cross the optical axis **185** beyond the image plane, at or near an alternate NP point **192B** that can be selected or preferred because it favors edge of field rays. However, as is also shown in FIG. 5C, such edge of field rays **172** need not cross the optical axis **185** at exactly the same point. Those differences, when translated back to object space **105**, translate into small differences in the parallax or perspective for imaged ray bundles or fans within or across an imaged FOV (e.g., a Core FOV **205**, as in FIG. 7) of a camera lens.

[0086] A second, adjacent camera lens **120B**, shown in FIG. 5D, can provide a similar performance, and image a fan of chief rays **170**, including ray fan **179C**, from within a Core FOV **205** with a vectorial projection of these chief rays converging within a corresponding low parallax volume **188B**. LP volumes **188A** and **188B** can overlap or be coincident, or be offset, depending on factors including the camera geometries and the seams between adjacent cameras, or lens system fabrication tolerances and compensators, or on whether the device center **196** is offset from the LP volumes **188**. The more overlapped or coincident these LP volumes **188** are, the more overlapped are the centers of perspective of the two lens systems. Ray Fan **179B** of camera lens **120A** and ray fan **179C** of camera lens **120B** are also nominally parallel to each other; e.g., there is no parallax error between them. However, even if the lens designs allow very little residual parallax errors at the FOV edges, fabrication variations between lens systems can increase the differences.

[0087] Analytically, the chief ray data from a real lens can also be expressed in terms of perspective error, including chromatic errors, as a function of field angle. Perspective error can then be analyzed as a position error at the image between two objects located at different distances or directions. Perspective errors can depend on the choice of COP location, the angle within the imaged FOV, and chromatic errors. For example, it can be useful to prioritize a COP so as to minimize green perspective errors. Perspective differences or parallax errors can be reduced by optimizing a chromatic axial position (Δz) or width within an LP volume **188** related to a center of perspective for one or more field angles within an imaged FOV. The center of perspective can also be graphed and analyzed as a family of curves, per color, of the Z (axial) intercept position (distance in mm) versus field angle. Alternately, to get a better idea of what a captured image will look like, the COP can be graphed and analyzed as a family of curves for a camera system, as a parallax error in image pixels, per color, versus field.

[0088] During the design of a camera lens systems, a goal can be to limit the parallax error to a few pixels or less for imaging within a Core FOV **205** (FIG. 7). Alternately, it can be preferable to particularly limit parallax errors in the peripheral fields, e.g., for the outer edges of a Core FOV and for an Extended FOV region (if provided). If the residual parallax errors for a camera are thus sufficiently small, then the parallax differences seen as a perspective error between two adjacent cameras near their shared seam **160**, or within a seam related region of extended FOV overlap imaging, can likewise be limited to several pixels or less (e.g., ≤ 3 -4 pixels). Depending on the lens design, device design, and application, it can be possible and preferable to reduce parallax errors for a lens system further, as measured by perspective error, to ≤ 0.5 pixel for an entire Core FOV, the peripheral fields, or both. If these residual parallax errors for each of two adjacent cameras are small enough, images can be acquired, cropped, and readily tiled, while compensating for or hiding image artifacts from any residual seams **160** or blind regions **165**.

[0089] In pursuing the design of a panoramic camera of the type of that of FIG. 1, but to enable an improved low-parallax multi-camera panoramic capture device (**300**), having multiple adjacent

cameras, the choices of lens optimization methods and parameters can be important. A camera lens **120**, or system of lens elements **135**, like that of FIG. 2A, can be used as a starting point. The camera lens has compressor lens element(s), and inner lens elements **140**, the latter of which can also be defined as consisting of a pre-stop wide angle lens group, and a post-stop eyepiece-like lens group. In designing such lenses to reduce parallax errors, it can be valuable to consider how a fan of paraxial to non-paraxial chief rays **125** (see FIG. 2A), or a fan of edge chief rays **170** (see FIG. 2B), or localized collections of edge of field rays **172** (see FIG. 5C) or **179 A,B** (see FIG. 5D) are imaged by a camera lens assembly. It is possible to optimize the lens design by using a set of merit function operands for a collection or set (e.g., 31 defined rays) of chief rays, but the optimization process can then become cumbersome. As an alternative, in pursuing the design of an improved low-parallax multi-camera panoramic capture device (**300**), it was determined that improved performance can also be obtained by using a reduced set of ray parameters or operands that emphasizes the transverse component of spherical aberration at the entrance pupil, or at a similar selected surface or location (e.g., at an offset NP point **192A** or **192B**) within an LP smudge volume **188** behind the lens system. Optimization for a transverse component of spherical aberration at an alternate non-paraxial entrance pupil can be accomplished by using merit function weightings that emphasize the non-paraxial chief rays.

[0090] As another aspect, in a low-parallax multi-camera panoramic capture device, the fans of chief rays **170** that are incident at or near a beveled edge of an outer lens element of a camera **120** (see FIG. 2B) should be parallel to a fan of chief rays **170** that are incident at or near an edge **132** of a beveled surface of the outer lens element of an adjacent camera (see FIG. 1). It is noted that an “edge” of an outer lens element **137** or compressor lens is a 3-dimensional structure (see FIG. 2B), that can have a flat edge cut through a glass thickness, and which is subject to fabrication tolerances of that lens element, the entire lens assembly, and housing **130**, and the adjacent seam **160** and its structures. The positional definition of where the beveled edges are cut into the outer lens element depends on factors including the material properties, front color, distortion, parallax correction, tolerances, and an extent of any extra extended FOV **215**. An outer lens element **137** becomes a faceted outer lens element when beveled edges **132** are cut into the lens, creating a set of polygonal shaped edges that nominally follow a polygonal pattern (e.g., pentagonal or hexagonal). [0091] A camera system **120** having an outer lens element with a polygonal shape that captures incident light from a polygonal shaped field of view can then form a polygonal shaped image at the image plane **150**, wherein the shape of the captured polygonal field of view nominally matches the shape of the polygonal outer lens element. The cut of these beveled edges for a given pair of adjacent cameras can affect both imaging and the optomechanical construction at or near the intervening seam **160**.

[0092] As another aspect, FIG. 5E depicts “front color”, which is a difference in the nominal ray paths by color versus field, as directed to an off axis or edge field point. Typically, for a given field point, the blue light rays are the furthest offset. As shown in FIG. 5E, the accepted blue ray **157** on a first lens element **137** is $\Delta X \approx 1$ mm further out than the accepted red ray **158** directed to the same image field point. If the lens element **137** is not large enough, then this blue light can be clipped or vignetted and a color shading artifact can occur at or near the edges of the imaged field. Front color can appear in captured image content as a narrow rainbow-like outline of the polygonal FOV or the polygonal edge of an outer compressor lens element **437** which acts as a field stop for the optical system. Localized color transmission differences that can cause front color related color shading artifacts near the image edges can be caused by differential vignetting at the beveled edges of the outer compressor lens element **137**, or from edge truncation at compressor lens elements (FIGS. **8A**, **9A**), or through the aperture stop **145**. During lens design optimization to provide an improved low-parallax camera lens (**320**), front color can be reduced (e.g., to $\Delta X \leq 0.5$ mm width) as part of the chromatic correction of the lens design, including by glass selection within the compressor lens group or the entire lens design, or as a trade-off in the correction of lateral color. The effect of front

color on captured images can also be reduced optomechanically, by designing an improved camera lens (320) to have an extended FOV 215 (FIG. 7), and also the opto-mechanics to push straight cut or beveled lens edges 132 at or beyond the edge of the extended FOV 215, so that any residual front color occurs outside the core FOV 220. The front color artifact can then be eliminated during an image cropping step during image processing. The impact of front color or lateral color can also be reduced by a spatially variant color correction during image processing. As another option, an improved camera lens (320, 920) can have a color dependent aperture at or near the aperture stop, that can, for example, provide a larger transmission aperture (diameter) for blue light than for red or green light.

[0093] Optical performance at or near the seams can be understood, in part, relative to distortion (FIG. 6) and a set of defined fields of view (FIG. 7). In particular, FIG. 7 depicts potential sets of fields of view for which potential image light can be collected by two adjacent cameras. As an example, a camera with a pentagonally shaped outer lens element, whether associated with a dodecahedron or truncated icosahedron or other polygonal lens camera assembly, with a seam 160 separating it from an adjacent lens or camera channel, can image an ideal FOV 200 that extends out to the vertices (60) or to the polygonal edges of the frustum or conical volume that the lens resides in. However, because of the various physical limitations that can occur at the seams, including the finite thicknesses of the lens housings, the physical aspects of the beveled lens element edges, mechanical wedge, and tolerances, a smaller core FOV 205 of transiting image light can actually be imaged. The coated clear aperture for the outer lens elements 137 should encompass at least the core FOV 205 with some margin (e.g., 0.5-1.0 mm). As the lens can be fabricated with AR coatings before beveling, the coatings can extend out to the seams. The core FOV 205 can be defined as the largest low parallax field of view that a given real camera 120 can image. Equivalently, the core FOV 205 can be defined as the sub-FOV of a camera channel whose boundaries are nominally parallel to the boundaries of its' polygonal cone (see FIGS. 5A and 5B). Ideally, with small seams 160, and proper control and calibration of FOV pointing, the nominal Core FOV 205 approaches or matches the ideal FOV 200 in size.

[0094] During a camera alignment and calibration process, a series of image fiducials 210 can be established along one or more of the edges of a core FOV 205 to aid with image processing and image tiling or mosaicing. The resulting gap between a core FOV 205 supported by a first camera and that supported by an adjacent camera can result in blind regions 165 (FIG. 5A, B). To compensate for the blind regions 165, and the associated loss of image content from a scene, the cameras can be designed to support an extended FOV 215, which can provide enough extra FOV to account for the seam width and tolerances, or an offset device center 196. As shown in FIG. 7, the extended FOV 215 can extend far enough to provide overlap 127 with an edge of the core FOV 205 of an adjacent camera, although the extended FOVs 215 can be larger yet. This limited image overlap can result in a modest amount of image resolution loss, parallax errors, and some complications in image processing as were previously discussed with respect to FIG. 3, but it can also help reduce the apparent width of seams and blind regions. However, if the extra overlap FOV is modest (e.g., $\leq 5\%$) and the residual parallax errors therein are small enough (e.g. ≤ 0.75 pixel perspective error), as provided by the present approach, then the image processing burden can be very modest. Image capture out to an extended FOV 215 can also be used to enable an interim capture step that supports camera calibration and image corrections during the operation of an improved panoramic multi-camera capture device 300. FIG. 7 also shows an inscribed circle within one of the FOV sets, corresponding to a subset of the core FOV 205, that is the common core FOV 220 that can be captured in all directions from that camera. The angular width of the common core FOV 220 can be useful as a quick reference for the image capacity of a camera. An alternate definition of the common core FOV 220 that is larger, to include the entire core FOV 205, can also be useful. The dashed line (225) extending from the common core FOV 220 or core FOV 205, to beyond the ideal FOV 200, to nominally include the extended FOV 215, represents a region in

which the lens design can support careful mapping of the chief or principal rays or control of spherical aberration of the entrance pupil, so as to enable low-parallax error imaging and easy tiling of images captured by adjacent cameras.

[0095] Across a seam **160** spanning the distance between two adjacent usable clear apertures between two adjacent cameras, to reduce parallax and improve image tiling, it can be advantageous if the image light is captured with substantial straightness, parallelism, and common spacing over a finite distance. The amount of FOV overlap needed to provide an extended FOV and limit blind regions can be determined by controlling the relative proximity of the entrance pupil (paraxial NP point) or an alternate preferred plane within a low parallax volume **188** (e.g., to emphasize peripheral rays) to the device center **196** (e.g., to the center of a dodecahedral shape). The amount of Extended FOV **215** is preferably 5% or less (e.g., $\leq 1.8^\circ$ additional field for a nominal Core FOV of 37.5°), such that a camera's peripheral fields are then, for example, ~ 0.85 - 1.05). If spacing constraints at the device center, and fabrication tolerances, are well managed, the extended FOV **215** can be reduced to $\leq 1\%$ additional field. Within an extended FOV **215**, parallax should be limited to the nominal system levels, while both image resolution and relative illumination remain satisfactory. The parallax optimization to reduce parallax errors can use either chief ray or pupil aberration constraints, and targeting optimization for a high FOV region (e.g., 0.85 - 1.0 field), or beyond that to include the extra camera overlap regions provided by an extended FOV **215** (e.g., FIG. 7, a fractional field range of ~ 0.85 - 1.05).

[0096] In addition, in enabling an improved low-parallax multi-camera panoramic capture device (**300**), with limited parallax error and improved image tiling, it can be valuable to control image distortion for image light transiting at or near the edges of the FOV, e.g., the peripheral fields, of the outer lens element. In geometrical optics, distortion is a deviation from a preferred condition (e.g., rectilinear projection) that straight lines in a scene remain straight in an image. It is a form of optical aberration, which describes how the light rays from a scene are mapped to the image plane. In general, in lens assemblies used for image capture, for human viewing it is advantageous to limit image distortion to a maximum of $\pm 2\%$. In the current application, for tiling or combining panoramic images from images captured by adjacent cameras, having a modest distortion of $\leq 2\%$ can also be useful. As a reference, in barrel distortion, the image magnification decreases with distance from the optical axis, and the apparent effect is that of an image which has been mapped around a sphere (or barrel). Fisheye lenses, which are often used to take hemispherical or panoramic views, typically have this type of distortion, as a way to map an infinitely wide object plane into a finite image area. Fisheye lens distortion (**251**) can be large (e.g., 15% at full field or 90° half width (HW)), as a deviation from f-theta distortion, although it is only a few percent for small fields (e.g., $\leq 30^\circ$ HW). As another example, in laser printing or scanning systems, f-theta imaging lenses are often used to print images with minimal banding artifacts and image processing corrections for pixel placement. In particular, F-theta lenses are designed with a barrel distortion that yields a nearly constant spot or pixel size, and a pixel positioning that is linear with field angle θ , ($h=f*\theta$).

[0097] Thus, improved low-parallax cameras **320** that capture half FOVs of ≤ 35 - 40° might have fisheye distortion **251**, as the distortion may be low enough. However, distortion can be optimized more advantageously for the design of improved camera lens assemblies for use in improved low-parallax multi-camera panoramic capture devices (**300**). As a first example, as shown in FIG. 6, it can be advantageous to provide camera lens assemblies with a localized nominal f-theta distortion **250A** at or near the edge of the imaged field. In an example, the image distortion **250** peaks at ~ 0.75 field at about 1% , and the lens design is not optimized to provide f-theta distortion **250** below ~ 0.85 field. However, during the lens design process, a merit function can be constrained to provide a nominally f-theta like distortion **250A** or an approximately flat distortion **250B**, for the imaged rays at or near the edge of the field, such as for peripheral fields spanning a fractional field range of ~ 0.9 - 1.0 . This range of high fields with f-theta type or flattened distortion correction includes the

fans of chief rays **170** or perimeter rays of FIG. 2B, including rays imaged through the corners or vertices **60**, such as those of a lens assembly with a hexagonal or pentagonal outer lens element **137**. Additionally, because of manufacturing tolerances and dynamic influences (e.g., temperature changes) that can apply to a camera **120**, including both lens elements **135** and a housing **130**, and to a collection of cameras **120** in a panoramic multi-camera capture device, it can be advantageous to extend the region of nominal f-theta or flattened distortion correction in peripheral fields to beyond the nominal full field (e.g., 0.85-1.05). This is shown in FIG. 6, where a region of reduced or flattened distortion extends beyond full field to ~1.05 field. In such a peripheral field range, it can be advantageous to limit the total distortion variation to $\leq 0.5\%$ or less. Controlling peripheral field distortion keeps the image “edges” straight in the adjacent pentagonal shaped regions. This can allow more efficient use of pixels when tiling images, and thus faster image processing.

[0098] The prior discussion treats distortion in a classical sense, as an image aberration at an image plane. However, in low-parallax cameras, this residual distortion is typically a tradeoff or nominal cancelation of contributions from the compressor lens elements (**137**, or **937** and **938** in FIG. 9A) versus those of the aggregate inner lens elements (**140**, or **940** in FIG. 9A). Importantly, the ray re-direction caused by the distortion contribution of the outer compressor lens element also affects both the imaged ray paths and the projected chief ray paths towards the low parallax volume. This in turn means that for the design of at least some low-parallax lenses, distortion optimization can affect parallax or edge of field NP point or center of perspective optimization.

[0099] The definitions of the peripheral fields or a fractional field range **225** of (e.g., ~0.85-1.05, or including $\leq 5\%$ extra field), in which parallax, distortion, relative illumination, resolution, and other performance factors can be carefully optimized to aid image tiling, can depend on the device and camera geometries. As an example, for hexagonal shaped lenses and fields, the lower end of the peripheral fields can be defined as ~0.83, and for pentagonal lenses, ~0.8. Although FIG. 7 was illustrated for a case with two adjacent pentagon-shaped outer lens elements and FOV sets, the approach of defining peripheral fields and Extended FOVs to support a small region of overlapped image capture, can be applied to multi-camera capture device designs with adjacent pentagonal and hexagonal cameras, or to adjacent hexagonal cameras, or to cameras with other polygonal shapes or with adjacent edges of any shape or contour generally.

[0100] For an Extended FOV **215** to be functionally useful, the nominal image formed onto an image sensor that corresponds to a core FOV **205** needs to underfill the used image area of the image sensor, by at least enough to allow an extended FOV **215** to also be imaged. This can be done to help account for real variations in fabricated lens assemblies from the ideal, or for the design having an offset device center **196**, as well as fabrication variations in assembling an improved low-parallax multi-camera panoramic capture device (**300**). But as is subsequently discussed, prudent mechanical design of the lens assemblies can impact both the imaged field of view of a given camera and the seams between the cameras, to limit mechanical displacements or wedge and help reduce parallax errors and FOV overlap or underlap. Likewise, tuning the image FOV (core FOV **205**) size and position with compensators or with fiducials and image centroid tracking and shape tracking can help. Taken together in some combination, optimization of distortion and low or zero parallax imaging over extended peripheral fields, careful mechanical design to limit and compensate for component and assembly variations, and the use of corrective fiducials or compensators, can provide a superior overall systems solution. As a result, a captured image from a camera can readily be cropped down to the nominal size and shape expected for the nominal core FOV **205**, and images from multiple cameras can then be mosaiced or tiled together to form a panoramic image, with reduced burdens on image post-processing. However, an extended FOV **215**, if needed, should provide enough extra angular width (e.g., $\square_{\text{sub.1}} \leq 5\%$ of the FOV) to match or exceed the expected wedge or tilt angle $\square_{\text{sub.2}}$, that can occur in the seams, $\square_{\text{sub.1}} \geq \square_{\text{sub.2}}$.

[0101] In designing an improved imaging lens (**320**, **920**) of the type that can be used in a low-

parallax panoramic multi-camera capture device (**100** or **300**), several first order parameters can be calculated so as to inform the design effort. A key parameter is the target size of the frustum or conical volume, based on the chosen polygonal configuration (lens size (FOV) and lens shape (e.g., pentagonal)) and the sensor package size. Other key parameters that can be estimated include the nominal location of the paraxial entrance pupil, the focal lengths of the compressor lens group and the wide-angle lens group, and the FOV seen by the wide-angle group.

[0102] But the design optimization for an improved camera lens (**320**, **920**) for use in an improved low-parallax panoramic multi-camera capture devices (**300**) also depends on how the numerous other lens attributes and performance metrics are prioritized. In particular, the relevant system parameters can include the control of parallax or the center of perspective (COP) error at the edges of an imaged field or for inner field locations or both, as optimized using fans of chief rays or spherical aberration of the entrance pupil). These parameters are closely linked with other key parameters including the width and positions of the “LP smudge” or volume **188**, the size of any center offset distance between the entrance pupil or LP smudge and the device center **196**, the target width of the gaps or seams, the extent of blind regions **165**, and the size of any marginal or extended FOV to provide overlap. The relevant performance metrics can include image resolution or MTF, distortion (particularly in the peripheral fields, and distortion of the first compressor lens element and of the compressor lens group), lateral color, relative illumination, front color, and color vignetting, telecentricity, and ghosting. Other relevant design variables can include mechanical and materials parameters such as the number of compressor lens elements, the configuration of the compressor lens group, the wide-angle lens group and eyepiece lens group, glass choices, the allowed maximum size of the first compressor or outer lens element, the sensor package size, the track length, the nominal distance from the image plane to the nearest prior lens element (e.g., working distance), the nominal distance from the image plane to the entrance pupil, the nominal distance from the image plane or the entrance pupil to the polygonal center or device center, manufacturing tolerances and limits, and the use of compensators.

[0103] While improving the optical design of the camera lens systems is important for enabling improved low-parallax panoramic multi-camera capture devices (**300**), improving the opto-mechanical design can be equivalently important. As suggested previously, the actual performance of a camera **120** can vary from the designed performance due to materials and fabrication variations amongst the individual lens elements **135** and the housing **130** and its constituent components, and the interactions thereof. As a result of such variations, the image quality (e.g., aberrations, including distortion), focal length (EFL) and magnification, working distance or track length, beam pointing or image location, and other attributes of a camera **120** can vary. These variations also mean that the assembly and performance of a given camera varies from that of another camera with nominally the identical opto-mechanical design. For example, the focal length of a set of nominally identical cameras can vary by $\pm 2\%$, which in turn will cause a similar variation in the lens magnifications and FOVs. This variation can be reduced or eliminated by designing improved camera lenses varifocally to include a focal length compensator, such as with a lens element whose axial position can be adjusted. Alternately, the cameras **120** can be designed such that nominal image from a nominal camera underfills the image sensor, enough so that the image from a camera with a large (e.g., $+2\%$) focal length lens also underfills a sensor, albeit with less margin. During calibration to determine a FOV, the EFL or magnification of a lens can be measured, and the sensor can also be aligned to be in focus for that lens. Image processing software can then be used to compensate the image for the lens variations, including compensating image size for magnification and distortion variations between lenses.

[0104] As previously discussed, an improved low-parallax camera lens can be optimized using a group of edge of field chief rays (**170**) or by limiting spherical aberration of the entrance pupil relative to a low parallax volume. But analysis or optimization of perspective errors can be another useful way to optimize the parallax errors for cameras used in an improved low-parallax panoramic

multi-camera capture device. Essentially, parallax (image perspective error (in pixels)) is related to spatial changes in the center of perspective, $\square$ COP. Analysis using perspective errors or the COP has the advantage of being more directly relatable to visible image artifacts, as compared to spherical aberration of the pupil or LP smudge measurements. Thus, consideration of perspective errors in captured images can be useful in defining target values for less intuitive parameters such as spherical aberration of the pupil that can be used directly in optimization. Likewise, it can also be useful during optimization to assess perspective errors for a single lens channel or camera objective, with emphasis on improving performance within a periphery of the captured FOV, or for providing an averaged performance of reduced errors across a larger captured FOV.

[0105] In the subsequent discussion, aspects of several alternate or improved imaging lenses **920** (FIGS. **8A**, **9A**, **10A**, **11**, and **12**) will be discussed in some detail, and compared to the prior design of FIGS. **2A**, **B**. As a first alternate lens, FIG. **8A** depicts a camera lens **920** with lens elements **935**, that is an improved version of the lens **120** of FIG. **2A** that can be used in an improved low-parallax multi-camera panoramic capture device (**300**). The FIG. **2A** and FIG. **8A** lens systems have a common lens form, and some similar structural traits, despite also being different in many design and performance details. As before, image light **975** from object space **905** is refracted and transmitted through a first lens element **937** to further inner lens elements and an aperture stop **945**, to converge to a focused image at F/2.0 at or near an image plane **950**, where an image sensor (not shown) is typically located. As this lens is designed for a dodecahedral system (12 lenses, excluding the support), the first lens element **937** nominally images a FOV with chief rays at 37.377 deg. at the vertices (vertex ray **970**), and 31.717 deg. at the mid-chords (mid edge ray **972**). The chief ray projections converge or point towards an LP smudge **992** which includes a paraxial entrance pupil. The lens elements can be defined as consisting of a compressor lens group (e.g., first lens element **937**), a pre-stop wide angle lens group **960** having two lens elements, including a deeply concave lens element **967**, and a post-stop eyepiece lens group **965** having four lens elements. This improved lens **920**, is similar to the camera lens **120** of FIG. **2A**, but it has been designed for further improvements or reductions in parallax error (or center of perspective) and front color.

[0106] A prescription for this camera lens is given in FIG. **8C-1**, with glass or material types, axial thicknesses, and surface radii identified by lens or lens surface. The lens has 7 lens elements, three of which have aspherical surfaces, and one of which (deeply concave lens element **967**) consists of an optical polymer (Zeonex E48R) with a conic surface. The lens assembly fits within a track length **980** of 50.0 mm, and images to a 3.9 mm total width at the sensor plane. Aspherical surface coefficients are also provided in FIG. **8C-2**. This lens was designed for color imaging with nominal wavelengths at 450, 587, and 656 nm to an image sensor with $\sim 2.2 \square\text{m}$ wide pixels. This camera lens **920** has a nominal focal length of 2.57 mm, an aperture stop diameter of 1.42 mm, and a low parallax point volume **992** with a paraxial entrance pupil, and a non-paraxial chief ray NP point located in proximity therein. An offset distance **925**, as shown in FIG. **8A**, can be equivalent to the entrance pupil to image plane distance (EPID), and in this case, is ≈ 16.8 mm behind or beyond the image plane **950**. The ratio of EPID to the focal length, or EPID/EFL, in this example, is $-16.8 \text{ mm}/2.57 \text{ mm} \approx -6.5:1$. The negative value of this ratio, and its' magnitude, is indicative of the extreme positioning of the entrance pupil behind the image plane **950**.

[0107] FIG. **8B** and FIG. **8D-G** depict some relevant performance parameters for the example lens design described in FIG. **8A** and FIG. **8C**. As an alternative to interpreting an illustration of a low parallax volume (e.g., FIG. **5C**), FIG. **8B** depicts a variation of center of perspective **900**, as an error or difference in image pixels versus field angle and color (R, G, B) for the optimized lens of FIG. **8A**. In this example, imaging of two objects, one at a 3 foot distance from an improved low-parallax multi-camera panoramic capture device (**300**) having camera lenses **320** of the design of FIG. **8A**, and the other at an "infinite" (∞) distance from the device, were analyzed. FIG. **8B** shows parallax errors of <1 pixel for all colors, from on axis, to nearly the edge of the field (e.g., to ~ 34

deg.). Parallax errors for blue light exceed 1.5 pixels out at the extreme field points (e.g., the vertices). However, most visible imaging systems, including the human visual system, and cameras using Bayer type color filter arrays, are desensitized to resolution type errors when imaging in blue light, relative to imaging red and green light. In general, providing parallax errors of ≤ 2 pixels from a camera, within its core FOV **205**, and particularly the peripheral fields thereof, and preferably also within a modest sized extended FOV **215**, can limit residual image artifacts to acceptable and hard to detect levels. But limiting perspective or parallax errors further, to sub-pixel levels (e.g., ≤ 0.5 pixel) for imaging within these FOVs, and particularly within the peripheral fields, for at least green light, is preferable. If the residual parallax errors between adjacent cameras are small enough, the captured images obtained from the core FOVs can be readily and quickly cropped and tiled together. Likewise, if the residual parallax errors within the extended FOVs that capture content in or near the seams are similarly small enough, and the two adjacent cameras are appropriately aligned to one another, then the overlapped captured image content by the two cameras can be quickly cropped or averaged and included in the output panoramic images.

[0108] In particular, FIG. **8D** depicts the resolution for this lens, as the MTF (modulation transfer function) performance, which is generally above 0.75 for most field positions out to 150 cy/mm. Generally, for many optical applications where the MTF versus field is harder to obtain than in this design example, an MTF ≥ 0.4 at the edge of the field is a reasonable target. FIG. **8E** depicts the astigmatism and field curvature focal shifts for this lens, which is $\leq \pm 0.025$ mm over the field. The astigmatic difference in focus for the tangential and sagittal meridians in this design causes a modest drop in MTF at or near the polygonal vertices. FIG. **8E** also depicts the modeled distortion performance for this lens, which is $\leq 1.0\%$ over nearly the entire field, except at the vertices. In this example, the distortion follows a non-monotonic s-shaped curve, due to the combined contributions of multiple orders (e.g., 3.sup.rd, 5.sup.th and 7.sup.th) of distortion. This lens is also nominally telecentric to the image plane, to $\leq 3.9^\circ$ for all field points and colors.

[0109] Because these camera lens systems (**920**) are adjacently abutted within an improved low-parallax multi-camera panoramic capture device (**300**), to then provide panoramic images, other parameters can matter in the design of these lenses, more so than with typical lens systems. FIG. **8F** depicts the relative illumination, which is $> 70\%$ across the entire field, including at or near edge of the field (~ 37 deg.). Relative Illumination (RI) is the combined effect of vignetting and radiometric roll-off in an imaging lens, and is generally given as a percentage of illumination at any point on an image plane, normalized to the position in the field (generally, the on-axis position) with maximum illumination. The graph of FIG. **8F** shows different curves for RGB relative illumination, in part because the Code V lens design program assumes the lens element surfaces have a simple MgF anti-reflection (AR) coating. But use of typical multi-layer AR coatings can raise these curves and reduce the color differences. Chromatic differences in vignetting through a lens system can also cause RI differences versus color. Increasing RI versus field helps reduce exposure and gain differences within a camera and between adjacent cameras, and thus makes the tiling of images easier. Preferably, a target level for RI is ≥ 0.50 at the extremes of the imaged FOV. On the other hand, extreme angles of incidence (AOIs) to lens element surfaces can increase Fresnel reflection losses, even with AR coatings, and reduce RI.

[0111] FIG. **8G** depicts “front color”, which is a difference in the nominal ray paths by color versus field, as directed to an off axis or edge field point. As shown in FIG. **8G**, the accepted blue ray on the outer surface of the first lens element **937** is ~ 1 mm further out than the accepted red ray directed to the same image field point. Stated another way, in this example, the width of front color near the edges of the outer lens element is ~ 1.0 mm. If that lens element is not big enough, then this blue light can be clipped or vignetted and a color shading artifact can occur at or near the edges of the imaged field. More broadly, localized color transmission differences that can cause front color related color shading artifacts near the image edges can be caused by differential vignetting at the beveled edges of the outer compressor lens element **937**, or from edge truncation at other

compressor lens elements **938** (FIG. **9A** and FIG. **10A**), or through the aperture stop **945**. As noted previously, with respect to FIG. **5E**, front color can appear in captured image content as a narrow rainbow-like outline of the polygonal edge of an outer compressor lens element **937**, which acts as a field stop for the optical system. As was also noted previously, front color can be reduced or compensated for by various means, including as part of the chromatic correction of the lens design, with appropriate glass selection or as a trade-off in the correction of lateral color. As front color can be thought of as an artifact of axial chromatic aberration at or near the entrance pupil, reducing the latter can help the former.

[0112] The example imaging lens **920** of FIGS. **8A-G** is a plausible lens for use in an improved low-parallax multi-camera panoramic capture device (**300**), but different design trade-offs can be made. For example, some MTF performance, relative illumination (RI) performance, lateral color performance, or telecentricity performance could be sacrificed to further improve front color or parallax error performance. Adding a second compressor lens element between the outer compressor lens element **937** and the deeply concave lens element **967** could be useful. As another option, a similar lens can also be designed with a greater use of optical polymers, but the color correction can then be harder to achieve.

[0113] Although this type of camera lens, or lens form, as exemplified by FIG. **8A**, with a compressor lens group or element (**937**), a pre-stop wide angle lens group **960**, and a post-stop eyepiece lens group **965**, may in entirety, or in part (lens groups **960** and **965**), visually resemble a fisheye lens, it is quite different. Unlike the present lens design (e.g., FIG. **8A**), a fisheye lens is an ultra-wide-angle lens that has heavily overcorrected spherical aberration of the pupil such that its entrance pupil is positioned near the front of the lens, in proximity to the first lens element. This pupil aberration also causes substantial shifts and rotations for the non-paraxial entrance pupils relative to the paraxial one. Such a lens is also reverse telephoto to provide a long back focal length, and a positive value for a ratio of the entrance pupil to image plane distance (EPID), divided by the lens focal length (EPID/EFL). A fisheye lens also provides a strong visual distortion that typically follows a monotonic curve (e.g., $H=f \square(f-\theta)$), that images with a characteristic convex non-rectilinear appearance. The typical fisheye lens captures a nominal 180° wide full FOV, although fisheye lenses that capture images with even larger FOVs (270-310°) have been described in literature. By comparison, the wide-angle cameras of the present approach, used in an improved low-parallax multi-camera panoramic capture device (**300**), are purposefully designed with low distortion, particularly at or near the edges of the imaged FOV, so as to ease image cropping and tiling. Also, the present cameras, while wide angle, typically capture image light from a significantly smaller FOV than do fisheye lenses. For example, a camera for a dodecahedral device nominally captures images from a full width FOV of ≈ 63 -75°. Whereas, an octahedral device would have cameras nominally capturing image light from a full width FOV of ≈ 71 -110° width, and an icosahedral device would have cameras nominally capturing image light from a full width FOV of ≈ 42 -75° width.

[0114] The compressor lens or lens group (**937**) redirects the incident light sharply towards the wide-angle lens group **960**. The first compressor lens refracts the incident light so that light rays heading in a direction towards a paraxial entrance pupil (**190** (FIG. **5**) or a chief ray's offset NP point (**192**) are positioned in a proper position behind the image plane, so that both lens elements and the sensor can be packaged within a projected conical volume or frustum. The steep ray refraction provided by the first compressor lens, or a compressor lens group, dramatically shifts the inner lens elements and image plane to be closer to the compressor elements, and thus makes the lenses total track length shorter overall. The use of multiple compressor lens elements in a lens group, rather than a single lens element, provides more optimization variables and the opportunities to improve imaging performance. For the example lens of FIG. **8A**, the incoming extreme chief ray at 37.377 deg. can be directed inwards at an angle of ~ 65 deg. relative to the optical axis **985**. The wide-angle lens group **960** receives this light and bends it sharply through the small aperture stop

945 at angle of ~ 33 deg. relative to the optical axis. The eyepiece lens group (**965**) can then present the transiting image light, across the imaged FOV, in a nominally telecentric manner to the image plane **950**.

[0115] While the internal combination of the pre-stop wide angle lens group **960** and the post-stop eyepiece lens group **965**, are not used as a stand-alone system for the present applications, if the compressor group **937** was removed, these two inner groups can also work together to form images at or near the image plane or sensor. In the optical designs of the camera lenses (**920**), these lens groups, and particularly the wide-angle lens group **960**, visually resembles a door peeper lens design. However, while this combination of two groups of lens elements again may visually appear similar to a fisheye or door-peeper type lens, they again do not image with fisheye type f-theta lens distortion (e.g., $H=f\theta$).

[0116] In particular, for the example lens of FIG. **8A**, the two inner groups of lens elements were not designed and optimized to work in isolation without compressor lens elements, and parameters such a resolution or chromatic correction can be sub-optimal. But when analyzed in isolation, without a compressor lens, the pre-stop wide angle lens group **960** and the post-stop eyepiece lens group **965** of the FIG. **8A** lens only needs a slight defocus correction. Notably, in this example, the pre-stop wide angle lens group **960** and the post-stop eyepiece lens group **965** do not provide F-theta distortion, but instead provide a nearly constant, approximately 0.0% distortion, out to $\sim 45^\circ$ FOV, and then the distortion grows rapidly with a negative or barrel distortion (e.g., to -20% at 65° FOV) thereafter. The compressor lens provides an approximately opposite or canceling distortion, so that the net distortion (FIG. **8E**) for the entire lens system is a few percent or less.

[0117] By comparison, the optical construction of the rear lens group (**965**), or sub-system, resembles that of an eyepiece, similar to those used as microscopic or telescopic eyepieces, but used in reverse, and without an eye being present. Eyepieces are optical systems where the entrance pupil is invariably located outside of the system. The entrance pupil of the eyepiece, where an eye would be located in a visual application, nominally overlaps with the plane where the aperture stop **945** is located. Likewise, the nominal input image plane in a visual application corresponds to the sensor plane (**950**) in the present application. The eyepiece lens group (**965**) was not designed to work with an eye, and thus does not satisfy the requirements for an actual eyepiece relative to eye relief, accommodation, FOV, and pupil size. But this eyepiece-like lens group solves a similar problem, and thus has a similar form to that of an eyepiece. Depending on the application, the optical design can more or less provide nominal optical performance similar to that of a more typical eyepiece.

[0118] In combination, a camera lens **920** comprises a compressor lens group (first lens element **937**), a wide-angle lens group, and an eyepiece lens group **965**, that has traits or aspects that are similar to other recognizable lens forms. Understanding these comparisons is useful to informing the lens design, tolerancing, mechanical design, alignment, and assembly of these lens systems. However, the overall camera lens **920** is unique in correcting or controlling parallax errors, which includes providing extreme positioning of the entrance pupil (or LP smudge **992**) well behind or beyond the image plane **950**.

[0119] FIG. **9A** depicts an alternate improved camera lens **920** with lens elements **935**, that is an enhanced version of the lens **120** of FIG. **2A** that can be used in an improved low-parallax multi-camera panoramic capture device (**300**). FIG. **9A** illustrates the overall lens on the left, and a zoomed in portion that illustrates the inner lens elements **940** in greater detail. This lens, which is also designed for a dodecahedral system, has lens elements **935** that includes both a compressor lens group **955** consisting of outer lens element **937** and compressor lens elements **938**, and inner lens elements **940**. In this design, compressor elements **938** are not quite combined as a cemented or air space doublet. As also shown in FIG. **9A**, inner lens elements **940** consists of a front wide-angle lens group **960** and a rear eyepiece like lens group **965**. Image light **975** from object space **905** is refracted and transmitted through a compressor lens group **955** having three lens elements,

such that chief rays at 37.377 deg. at the vertices (vertex ray **970**) are redirected at a steep angle of ~80 deg. towards the optical axis **985**. The front wide-angle group **960** comprises two lens elements, which direct the transiting image light through the aperture stop **945**. The eyepiece lens group **965**, which has five lens elements, redirects the transiting image light coming from the aperture stop **945** to provide image light telecentrically at F/2.8 to an image sensor at an image plane **950**. As this lens is designed for a dodecahedral system, the first lens element **937** nominally accepts image light for a FOV width of 31.717 deg. at the mid-chords (mid edge ray **972**). The chief ray projections converge or point towards an LP smudge **992** which includes a paraxially defined entrance pupil.

[0120] This improved lens **920**, is similar to the improved camera lens **920** of FIG. **8A**, but it has been designed for a more demanding set of conditions relative to parallax correction, a larger image size (4.3 mm wide), and a further removed entrance pupil to provide more room for use of a larger sensor board. The last requirement increases the distance between the image sensor plane and the entrance pupil or low parallax volume **992**. In particular, the focal length is larger (5.64 mm) so as to project the image onto a large sensor. As shown in FIG. **9B**, which shows a portion of the eyepiece lens group, within the LP smudge **992**, there are several potentially useful planes or locations of reference, including the paraxial entrance pupil **990**, a location of a center of perspective **902**, locations for non-paraxial chief ray NP points, or a location of a circle of least confusion **995** where the LP smudge or parallax volume has a minimal size in the plane tangent to the optical axis. The entrance pupil **990** is a good reference as it is readily calculated from a common first order optics equation. The axial location of a center of perspective **902** is also a good reference as it is directly relatable to perceived image quality. While the distance from the image plane **950** to any of these locations can be used as a reference, an offset distance **925** to the entrance pupil **990** can be preferred. In this example (FIG. **9B**), the entrance pupil **990** is located ~30 mm behind the image plane **950**, for a negative entrance pupil distance to focal length ratio, EPID/EFL=-5.3:1. Depending on how it is measured, the LP smudge **992** can have an axial width **997** of ≤ 2 mm.

[0121] FIG. **10A** and FIG. **10B** depict a second alternate camera lens **920** with lens elements **935**, with a zoomed in view of the inner lens elements **940**. This camera lens **920** is an improved version of the lens **120** of FIG. **2A** that can be used in an improved low-parallax multi-camera panoramic capture device (**300**). This lens, which is also designed for a dodecahedral system, includes lens elements **935** that consists of a compressor lens group **955** and inner lens elements **940**.

Compressor lens group **955** consists of an outer lens element **937**, and compressor lens elements **938** that are provided in combination as a cemented doublet. As also shown in FIG. **10A**, inner lens elements **940** consists of a wide-angle lens group **960** and an eyepiece like lens group **965**. Image light **975** from object space **905** is refracted and transmitted through a compressor lens group **955** having three lens elements, such that chief rays at 37.377 deg. at the vertices (vertex ray **970**) are redirected at a steep angle of ~75 deg. towards the optical axis **985**. The wide-angle group **960** comprises three lens elements, which directs the transiting image light through aperture stop **945**. The eyepiece lens group **965**, which has five lens elements, redirects the transiting image light coming from the aperture stop **945** to provide image light telecentrically at F/2.8 to an image sensor at an image plane **950**. As this lens is designed for a dodecahedral system, the first lens element **937** nominally accepts image light for a FOV width of 31.717 deg. at the mid-chords (mid edge ray **972**). The chief ray projections converge or point towards an LP smudge **992** which includes a paraxial entrance pupil. As shown in FIG. **10B**, the lens has a track length **980**, from the axial vertex to the image plane **950**, of 78.5 mm.

[0122] This improved lens **920**, is similar to the improved camera lens **920** of FIG. **9A**, but it has been designed for an alternate combination of conditions. Although the image size is a bit smaller (3.7 mm wide), the parallax and front color are better corrected, and the entrance pupil is yet further removed to provide yet more room for the sensor board, wiring, and cooling hardware. In

particular, the lens provides a smaller focal length (4.9 mm), and a larger offset distance **925** (~35.5 mm) to the entrance pupil or LP smudge **992** behind or beyond the image plane **950**, at a relative ratio of -7.25:1.

[0123] Even though the camera lens of FIGS. **10A,B** has a similar construction to that of FIG. **8A**, with large compressor optics, a wide-angle group **960**, and an eyepiece group **965**, the details of the lens constructions, relative to the number and shape of lens elements, are visually quite different. The actual lens performance is also similar, but not identical. As an example, FIG. **10C** depicts the MTF performance of this second alternate camera lens **920**, which is ~0.75 as an average of all fields at 100 cy/mm. As another example, FIG. **10D** depicts the relative illumination, which is ~0.57 at full field (~37 deg.). While this performance is less than that of the FIG. **8F** example, it is still high enough to significantly reduce the amount of exposure corrections needed from a camera lens system to another adjacent abutting camera lens system (e.g., see FIG. **9**). In general, it is preferable to design the camera lenses **920** for use in an improved low-parallax multi-camera panoramic capture device (**300**) at the edge of the imaged FOV to have an $RI \geq 0.5$ and even more preferably to have $RI \geq 0.75$.

[0124] FIG. **11** depicts a third alternate camera lens **920** with lens elements **935**, including 4 optical polymer lens elements, that is an improved version of the lens **120** of FIG. **2A** that can be used in an improved low-parallax multi-camera panoramic capture device (**300**). This lens which is designed for a system that is shaped as a regular icosahedron (20 lenses, excluding the support) with triangular sides or lens faces, includes lens elements **935** that consist of a compressor lens group **955** and inner lens elements **940**. As also shown in FIG. **11**, inner lens elements **940** consist of a front wide-angle lens group **960** and a rear eyepiece like lens group **965**, having several aspheric or conic surfaces. Image light from object space **905** is refracted and transmitted through a compressor lens group **955** having a single lens element, such that chief rays at 37.4 deg. at the vertices (vertex ray **970**) are redirected at a steep angle of ~60 deg. towards the optical axis **985**. The front wide-angle group **960** comprises two lens elements, which directs the transiting image light through the aperture stop **945**. The eyepiece lens group **965**, which has four lens elements, redirects the transiting image light coming from the aperture stop **945** to provide image light at F/2 to an image sensor at an image plane **950**. The chief ray projections converge or point towards an LP smudge **992** which includes a paraxial entrance pupil.

[0125] FIG. **12** depicts a fourth alternate camera lens **920** with lens elements **935**, that is an improved version of the lens **120** of FIG. **2A** that can capture light in a 110° FOV, for use in an improved low-parallax multi-camera panoramic capture device (**300**) having an octahedral (8 lenses, excluding the support) construction. This lens includes lens elements **935**, which is also designed for a system that is shaped as an octahedron with triangular sides or lens faces, and that consists of a compressor lens group **955** and inner lens elements **940**. As also shown in FIG. **12**, inner lens elements **940** consists of a front wide-angle lens group **960** and a rear eyepiece like lens group **965**, having several aspheric surfaces. Image light from object space **905** is refracted and transmitted through a compressor lens group **955** having a single lens element, such that chief rays at 55 deg. at the vertices (vertex ray **970**) are redirected at a steep angle of ~80 deg. towards the optical axis **985**. The wide-angle group **960** comprises two lens elements, which directs the transiting image light through aperture stop **945**. The eyepiece lens group **965**, which has four lens elements, redirects the transiting image light coming from the aperture stop **945** to provide image light at F/2 to an image sensor at an image plane **950**. The chief ray projections converge or point towards an LP smudge **992** which includes a paraxial entrance pupil. Because of the larger FOV, the offset distance **925** between the LP smudge **992** and the image plane **950** tends to be smaller than in other systems.

[0126] The alternate example camera lens **920** of FIG. **12** was designed for potential use in a low-end surveillance or smart phone market. It can use several lens elements that can potentially be fabricated using optical polymers, such as the last element in the eyepiece lens group **965**, located

just before the cover glass of the image sensor **950**. This lens element has a size and aspherical shape similar to elements used in cell phone cameras. This lens has a 1.98 mm focal length and provides image light at F/2 to the image plane **950**.

[0127] Considered as a set, the improved camera lens systems **920** of FIGS. **8A**, **9A**, **10A**, **11**, and **12**, provide various examples of a lens form with a common feature set, consisting of an initial compressor lens group which bends the light sharply towards the optical axis, a physically much smaller wide angle lens group which redirects the light into the aperture stop, and an eyepiece-like lens group which directs and focuses the transiting image light to an image plane. The requirement to reduce parallax or perspective errors, while enabling multiple polygonal shaped cameras to be adjacently abutted to form a larger improved low-parallax multi-camera panoramic capture device (**300**) brings about an extreme lens form, where lens elements in the compressor lens group can be rather large (e.g., 80-120 mm in diameter), while typically at least some lens elements in the wide-angle and eyepiece lens groups are simultaneously rather small (e.g., 5-10 mm in diameter). In these type of lens designs, the first compressor lens element or outermost lens element **937**, and adjacent outer lens elements of adjacent lens systems, can alternately be part of a contiguous faceted dome or shell. It is also typical that several (e.g., 2-4) of the lens element surfaces have aspheric or conic surface profiles, so as to bend or direct light rays transiting near the edges of the lens elements differently than those transiting near the center or optical axis. Typically, the wide-angle lens group **960** also has a lens element (**967**) with a deeply concave surface. In some cases, during optimization, that surface can want to become hyper-hemispherical, although to improve element manufacturability, such profiles are preferably avoided. Another measure of the extreme characteristics of this lens form, is the offset distance of the paraxial entrance pupil (or similarly, the LP smudge) behind or beyond the image plane. Unlike typical lenses, the entrance pupil is not in front of the image plane but is instead pushed far behind or beyond it. This is highlighted by the negative entrance pupil to image plane distance/focal length ratio, EPID/EFL, which can range from -2:1 to -10:1, but which is typically $\geq -4:1$ in value.

[0128] Optimization of the size, position, and characteristics of the LP smudge or low parallax volume **992**, as depicted in exemplary detail in FIG. **5C**, impacts the performance and design of the improved camera lens systems **920**, as illustrated in FIGS. **8A**, **9A**, **10A**, **11**, and **12**. The low parallax volume optimization is heavily impacted by the merit function parameters and weightings on chief rays for both spherical aberration of the entrance pupil and axial or longitudinal chromatic aberration of the entrance pupil. Lens element and lens barrel fabrication tolerances can also impact the size and positioning of this volume, or equivalently, the amount of residual parallax error, provided by the lens. Thus, even though these lenses can be considered to have an extreme form, optimization can help desensitize the designs to fabrication errors, and provide insights on how and where to provide corrective adjustments or compensators.

[0129] As noted previously, the distortion of the compressor lens or compressor lens group (**955**) tends to cancel the combined distortion of the inner lens groups (**940**). The high magnitude distortion from the compressor lens group is useful in providing the ray bending to control parallax and to position the paraxial and non-paraxial “NP” points or low parallax volume beyond or behind the image plane. But then the inner groups of lens elements need to provide a large, approximately canceling distortion, so as to yield a small residual across the imaged FOV. As the shapes and magnitudes of the individual distortions of the compressor lens group, and that of the combined inner groups of lens elements, can both vary dramatically amongst the lens designs of this type, then the shape of the overall distortion can also vary dramatically, even though overall distortion has a magnitude of $<2\%$. Thus, the distortion at mid or peripheral fields can be locally optimized and have a curved shape (e.g., like FIG. **8E**) or a more flattened shape (e.g., like distortion **250B** in FIG. **6**). As another factor, it is noted that the distortion optimization trade-off between the compressor lens element and the inner lens elements can also affect the chief ray pointing from the first compressor lens element **937**, and thus it can also affect parallax correction, as represented by

the locations of the off axis or non-paraxial chief ray NP points within the low parallax volume. [0130] In designing lens systems of this type for visible applications, it can be rather useful to use high index, low to mid dispersion optical materials, such as Ohara S-LAH53 or SLAL-18, including particularly for the compressor lens elements. As another option, the optical ceramic, Alon, from Surmet Corporation of Burlington, MA, has comparable refractive indices to these materials, but even less dispersion, which can make it quite useful in designing these lenses. It can also be useful to use optical polymers or plastics in these lens designs, particularly to reduce cost and weight, but also for other reasons. The compressor lens elements, and particularly the first or outermost compressor lens element **937** can be a good candidate for a glass to polymer substitution, as it can be so large, and is subject to complex edge beveling. High refractive index optical polymers, such as OKP4 from Osaka Gas Chemicals, or EP5000 from Mitsubishi Gas Chemical, can be particularly useful for such purposes. Likewise, it can be beneficial to use an optical polymer for the deeply concave lens element **967** (such as Zeonex E48R), relative to fabricating surfaces with extreme hemispheric or conic profiles. Unfortunately, optical polymers have a much more limited range of optical properties than do optical glasses, and the high refractive index polymers have both lower refractive indices and more dispersion than do the glasses, which can constrain the optical designs or performance.

[0131] Depending on priorities, these lens systems can be optimized further, and the different variations of the lens form may be individually better suited for different markets or applications. In general, the outermost lens element, or first compressor lens element, has used Ohara SLAH52, SLAH53, and SLAH63 glasses (or equivalent glasses from Schott Glass (e.g., N-LAF36 and N-LASF43)), which are high index, low dispersion flint glasses with visible spectra refractive indices $n \sim 1.80$ and an Abbe number $V_d \sim 41.5$. It should be understood that other optical materials can be used for the lens elements in the camera lenses **520** generally, including for the compressor lens elements. For example, use of a high index, lower dispersion, mid-crown glass like Ohara SLAL-18 can be helpful for color correction. As another example, lens elements can also be made from optical ceramics such as Alon ($n \sim 1.79$, $V_d \sim 57-72$) or Spinel, which are extremely durable materials, similar to sapphire, but with excellent optical transparency, low dispersion, and a controllably modifiable isotropic crystalline structure. A lens design, using optical polymers for at least some of the large compressor lens elements and using high index, low dispersion flint glasses, or Alon or Spinel, for at least some of the small inner lens elements can also be advantageous. It should also be understood that the camera lenses of the present approach can also be designed with optical elements that consist of, or include, refractive, gradient index, glass or optical polymer, reflective, aspheric or free-form, Kinoform, fresnel, diffractive or holographic, sub-wavelength or metasurface, optical properties. These lens systems can also be designed with achromatic or apochromatic color correction, or with thermal defocus desensitization.

[0132] Another aspect of the design of these improved camera lenses **920**, for use in an improved low-parallax multi-camera panoramic capture device (**300**), relates to their fit within the larger polygonal (e.g., dodecahedron, icosahedron, Goldberg polyhedra) device. As shown in FIG. **10B**, the camera **920** can be positioned at an offset distance **925** in front of the image plane **950**. The exact distance **925**, relative to a location within a width **997** of the LP smudge **992**, depends on the location to which it is calculated or measured (e.g., the paraxial entrance pupil to the image plane distance (EPID), or the distance from the plane with a non-paraxial chief ray NP point to the image plane **950**), and the color or wavelengths (e.g., green) used. The device center **910** (e.g., the polygonal center) can also be located within the width **997** of the LP smudge **992**, or it can be located outside the smudge width or volume, at a center offset distance **915**. In some device designs, it can be particularly useful to provide an axial center offset distance **915**, such that the LP smudge **992** or entrance pupil **990** is positioned at an axial distance away from the device center **910**. Designing in a center offset distance **915** (e.g., 1-4 mm) can provide extra space for the power, communications, or cooling connections, cables, and for mechanical supports for the sensor

package **952**, or for other structural members. In the example of the camera system **920** depicted in FIG. **10B**, the improved low-parallax multi-camera panoramic capture device (**300**) has a dodecahedral geometry, and the device center **910** is the nominal center of the dodecahedral polygonal structure. This offset distance must be determined during the process of designing the cameras **920** and overall device **300**, as it interacts with the optimization of the lens near the edges of the FOV. Thus this optimization depends on, or interacts with, the budget for an extended FOV **215**, distortion correction, the control of front color, the optimization of reduce parallax or for the extent of the LP smudge **992**, and the sizing of the lens elements (particularly for the compressor lens group **955**).

[0133] In the subsequent discussion, issues around mounting or assembling improved low parallax camera lenses **320** for use in an improved low parallax multi-camera panoramic capture device (**300**) will be discussed. These improved camera lenses **320** can be based on any of the various designs for improved imaging or camera lenses **920** discussed relative to FIGS. **8A**, **9A**, **10A**, **11**, and **12**), or the lens of FIG. **2A**, or variants thereof.

[0134] FIG. **13** depicts a cross-sectional view of an improved opto-mechanics construction for a low parallax multi-camera panoramic image capture device **300**, and a 3D view of a camera channel thereof. Fabrication variations for individual cameras, and the opto-mechanical interactions between them, can have significant impact on the design and performance of such a multi-camera capture device **300**. During the initial assembly of a multi-camera capture device **300**, the interaction of tolerances and mechanical wedge between a housing **430** of a first camera **320** and a housing **430** of a second adjacent and abutting camera **320** can affect the seams **400**, the pointing of the Core FOV **205** or Extended FOV **215** of individual cameras (causing FOV overlap or underlap), and can thus effect a FOV captured by each of the cameras. Furthermore, co-alignment mounting stresses imparted to adjacent cameras by the close proximity mounting can physically distort one or more of the camera housings, and then potentially also distort or skew the optical imaging function of the camera lenses. As a result, a camera lens system can provide an image to the sensor that is shifted or rotated or tilted out of plane. These potential problems, or the risk of their occurrence, can be exacerbated by environmental influences, such as asymmetrical thermal loading or substantial asymmetrical optical loading.

[0135] To counter such issues, an improved multi-camera capture device **300**, as shown in FIG. **13**, can include features to provide kinematic type mounting of individual cameras **320** or objective lenses. In particular, FIG. **13** depicts two views of a dodecahedron multi-camera capture device **300**, including a partial cross-section in which **11** pentagonal cameras **320** are mounted to a central support **325** that occupies the nominal position of a twelfth potential camera channel. Each camera **320** has a separate base lens assembly or housing **430** that consists of a lens mount which mounts the compressor lens (**437**) while also mounting the inner lens elements **440** that together comprise a base lens assembly. Although for each camera **320**, the lens elements and housings **430** fit within the nominal conical space or volume, they need not nominally fill that space. Indeed, the abrupt ray bending provided by the compressor lens elements can mean that the inner lens elements **440** and their housings or barrels underfill the available space, and the overall lens housings **430** can taper further inwards, potentially leaving an open inner volume **390** between adjacent lens assemblies.

[0136] The housings **430** or base lens assemblies of FIG. **13** also include a turned section, that can be machined on a CNC multi axis (5-axis) machine, and that mates with a tripod-like channel centering hub **330**. The channel centering hubs **330** can be entirely turned on a lathe except for the pentagonal flange, which is completed in a finish operation after the lathe. Being turned on a lathe means that exceptional concentricity and runout can be achieved, helping with the ultimate alignment of the channel. The housing **430** mates with the inside diameter of the channel centering hub **330** which is a key part of a central mount mechanical assembly that is designed to have a fit with it that ranges from a slip fit to a light interference fit, so as to ensure axial alignment without significant variations due to gap tolerances. This same fit reduces perpendicularity errors with

respect to the channel axis.

[0137] The tripods or channel centering hubs **330** also include a turned section or ball pivot **340** that mates with a socket **345** of a spherical socket array **346** provided on the central support **325**. In this system, the camera **320** located in the polar position, opposite the central support **325**, is a rigidly placed reference channel. The center support **325** consists of a cylindrically shaped post with a ball on the top. The geometries for the center support **325** and tripods or centering hubs **330** can be designed to provide more space for power and communications cables, cooling lines, and mechanisms to secure the lens housings **430** or cables. In this example, the ball contains sockets **345**, each of which can receive a ball pivot **340**. The ball pivots **340** are at the ends of extended pins or ball pivot arms **342**. Although this ball and socket portion of the center support **325** mount can be expensive to machine, given the precision expected with respect to the position and depth of the sockets, the advantages are that centerline pointing is controlled, while there is only a single part per device **300** that demands exceptional precision. Whereas, each of the camera channels **320** may be machined with less precision, which eases both the fabrication and replacement costs.

[0138] The individual camera lens housings **430** of FIG. **13** can also be provided with external or outside channel to channel datums **335**, located midway along the pentagonal sides. Each of these channel-to-channel datums **335** can comprise two parallel convexly curved slightly protruding bars that are separated by an intervening groove. These datums are designed to provide both single point or localized kinematic contacts or interactions between lens housings, such that the datum features interweave in such a way that only one part or housing will dominate in terms of tolerance. Since they are interwoven, only the variation of one part will influence the distance between each camera channel, and thus influence the angle between the channels. In particular, if one datum **335** is larger it will dominate because the other will not make contact. Thus, only one tolerance contributes for two parts. That the channel to channel datums **335** are interwoven from one camera **320** to another, also limits lateral movement between mating (pentagonal) faces or sides, while allowing limited angular movement of the lens housings **430**.

[0139] Individually, and in aggregate, the interactions between camera lens housings **430** or base lens assembly's limits mechanical wedge or channel pointing errors (roll, pitch and yaw) between cameras due to both the ball and socket arrangement and the datum features (**335**). Each camera channel assembly works together with its neighbors to limit channel pointing error. The portion of the base lens assembly (**430**) that holds the outer lens element **437** or compressor lens also has internal functional datums that can locate the compressor lens perpendicular relative, to the optical or mechanical channel axis, and it has additional internal datum features that limit axial misalignment. As will be discussed in subsequent detail with respect to FIG. **16**, an additional set of alignment features provided on the ground edges of the compressor lens can function as a datum and interact with these internal datum features to limit pentagonal rotation about the channel axis as well.

[0140] The use of the alignment features depicted in FIG. **13**, and particularly the ball pivot and socket datums (**350** and **356**) and the channel to channel datum features **335** reduces the risks of rotation, pivoting, or splay from one camera channel (**320**) to another. Thus, these features also help enable the seams **400** between cameras **320** to have more consistent thicknesses, with respect to their nominal design values, than may have happened otherwise. The use of the internal features within the lens housing (e.g., compensators, adjustments screws, and shims) and external features between lens housings (e.g., channel to channel datums, ball and socket datums, and a channel loading support) help control Core FOV or Extended FOV pointing, so that one camera channel can be aligned to another adjacent channel. The combined use of channel to channel datums, ball and socket datums (FIG. **14**), and a channel loading support (FIG. **17**) can also help desensitize the device to mechanical or thermal loads.

[0141] FIG. **14** depicts a further design option for improving the opto-mechanical design of the outer lens elements **437**, or compressor lens elements, in proximity to the seams **400** in an

improved multi-camera panoramic image capture device **300**. In particular, the edges of an outer lens element **437** can protrude beyond the upper or outer edge of a lens housing **430** such that two adjacent outer lens elements **437** of adjacent cameras **320** can be in near contact at the seams **400**. If these outer lens elements **437** are fabricated at least in part with a somewhat compliant material, then some amount of actual physical contact can be allowed. If, however, these outer lens elements **437** are fabricated with a brittle material, such as glass, then greater care is required.

[0142] In the configuration of FIG. **14**, two adjacent outer lens elements **437** that have protruding edges are in near contact at the seams **400**. In a preferred design approach, the opposing edges are provided with a stepped edge angle **365** or structure. In the outermost portion, the two lenses and housing can provide a parallel seam **400** with a width of 1.0 mm or smaller.

[0143] In the innermost portion, where the inner beveled edges **370** of the adjacent outer lens elements **437** approach each other, the lens housing **430**, channel to channel datums **335**, and flat surface datums **670** can be provided. The edge of each outer lens element **437** then has a stepped edge groove **380** that can be filled with a compliant adhesive. Above that, along the edges **432** or seams **400**, spanning the outermost edge portion of the adjacent outer lens elements **437**, these lens elements can be nearly abutting, and separated by a gap that can be only 0.5 mm wide, or smaller. In practice, the optimization of the seam width can depend on how brittle or compliant the lens material is (glass or polymer), the flexibility of a seam filling adhesive, the use of other protective measures, and the application.

[0144] FIG. **17** provides greater detail on how these and other features can be used during alignment and assembly. In particular, FIG. **17** depicts a cross-sectional view of a portion of an improved multi-camera panoramic image capture device **300** of the present approach, in which 11 camera channels **320** are attached to a post or vertical central support **325** through which both wiring and cooling can be provided (see FIG. **15** for more details). FIG. **17** in particular depicts a design for the opto-mechanical hardware, albeit assembled without the lens elements (see FIGS. **13** and **15** for illustrations of opto-mechanical hardware including lens elements). FIG. **18A** depicts key elements or components used in this example alignment and assembly approach in yet greater detail.

[0145] As shown in FIG. **17**, a top camera **320** can be identified as a primary alignment channel **610**, while all other camera channels **320** shown are identified as secondary channels **615**. In particular, for the primary channel, the ball pivot arm **342** with ball pivot **340** of the channel centering hub **330**, mates into a ball socket datum **356** of the ball socket array **346**, where it can interact with a locking axial retention pin **348** that slips into an indent between the ball pivot arm **342** and the ball pivot **340**, to stop Z-axis (vertical) translation. An anti-rotation keying pin **349** is press fit into a localized hole provided in the ball pivot arm, so as to stop rotation of the primary camera channel **610** about the z-axis. The ball pivot arm **342** of the primary camera channel **610** has a light press fit into its ball socket datum **356**, so as to prevent rotations about the X and Y axes.

[0146] Alternately, the ball sockets **345** can have a latch mechanism (not shown) to load a ball pivot **340** against a ball socket datum **356**, as a means to provide both a loading force and a mechanism that has reduced sensitivity to a mechanical loading force being applied to the camera or the device. For example, a latching mechanism can have latches actuated with linkage assemblies. As compared to using a retention pin, a latching mechanism can be more compliant, robust, and reliable, if a camera channel **320** or the device **300** are impacted by a loading force. It is also noted that the lens housings **430** can also have identifying marks to ease alignment of a housing with adjacent lens housings.

[0147] As previously shown in FIG. **13**, and shown in yet greater detail in FIG. **18A**, each camera channel **320** has a pair of peripheral channel to channel datums **335** on each face, that are small (localized) and nominally centered along the faces or sides of the lens housing. For a pentagonal lens, each of the five sides would have a pair of datums (**335**). These datums are provided with curved surfaces **373** (shown in bold, and extended, for emphasis) so as to enable point to point type

contact (375). In particular, each pair of channel to channel datums 335 has side faces which act as lateral datums between camera channels. They are curved to accommodate relative angular movement between channels. Each side datum only has an effective single point of contact 375 where each of their mating datum radial surfaces meet. Since the radii are very large, the datum surfaces approach a straight line near the intended single point of contact. Thus, any shifting of the camera channels relative to each other that could result in an angular or centerline offset can then only have a minute effect on the relative lateral offset of a given camera channel.

[0148] As further shown in FIG. 17, an improved multi-camera panoramic image capture device 300 can also have a channel loading support 630 that biases all secondary channels 615 against the primary channel 610. The channel loading support 630 employs peripheral datum pairs that are nominally identical to the channel to channel datums 335 used on each channel. The channel loading support 630 can also have a spring element 635 to facilitate loading of the secondary channels 615 against the primary channel 610. The channel loading support 630 can also utilize a key feature for anti-rotation, although using a keyed support may impart superfluous constraints. [0149] The primary alignment channel 610 (see FIG. 17) is aligned and locked into place when its ball pivot arm 342 and ball pivot 340 are engaged with the pins (348, 349) of the channel centering hub 330. The secondary camera channels 615 are then added, and are loosely aligned with their ball pivots 340 fit against datums within the sockets of the channel centering hub 330. When the channel loading support 330 is added, at the seams 400, it nudges the secondary channels 615 both against the primary channel 610 and each other. The relative size of the ball socket 345 to the ball pivots 340 of the secondary channels are provided so that the secondary channels 615 are constrained by the ball socket 345 in only the Z-direction.

[0150] Then at the seams 400, for any two faces of adjacent camera channels (610, 615) to be parallel to each other, the configuration requires that at least three camera channels have their peripheral channel to channel datums 335, with points of contact 375, in contact with opposing channel to channel datums 335 with their points of contact 375. This effectively constrains the secondary channels 615 for three degrees of freedom (DOF). It is recognized that applying traditional kinematic mechanical design principles, in which motion and constraints in 6 DOF can be precisely limited with little crosstalk, can be tricky in a device that can intrinsically locate a multitude of complex polygonal faces (e.g., hexagonal or pentagonal) in close proximity. In a case where more than 3 camera channels are in close proximity, if not in contact, with each other, the camera channels could become over constrained. Because the system can then be only quasi-kinematic, mechanical stresses and strains could then cause component mis-alignment or damage. These potential problems can be overcome by a variety of methods. As one example, the relatively small size and centered locations of the channel to channel datums 335 can limit any angular or spatial misalignment to relatively small amounts. As another example, a compliant material such as RTV can be applied in the seams 400 to absorb some stresses or strains, while the lens housings 430 can be designed and fabricated to be sufficiently rigid to resist deformations, mis-alignments, or damage. Seam width variations, including dynamic ones caused by thermal or mechanical loads, can also be compensated by providing improved cameras 320 that capture image light 415 with an extended FOV 215.

[0151] As shown in FIG. 18A, a point of contact can occur along a line across the radial surface of a datum. Thus, each channel to channel datum 335 makes “line” contact with an adjacent set of channel to channel datums 335 if the mating surfaces are perfectly parallel. If not, only point contact is made. Typically, the peripheral channel to channel datums 335 only contact an opposing face at one point. For each channel to channel datum pair, only a single datum is likely to contact due to dimensional variations. This is an asset of this design approach, since only a single datum can influence the gap or seam 400 between adjacent camera channels. This approach can be used in various designs for an improved multi-camera panoramic image capture device 300, including a device with a “soccer ball” or truncated icosahedron (FIG. 4) geometry, where the primary

alignment (camera) channel can be an external hexagonally shaped lens. This approach can also be used in the construction of “hemispherical” devices.

[0152] FIG. **18B-1** depicts an across seam cross-sectional view of an alternate or refined version of the construction of lens housings **430**, and their interface near a seam **400**, to that shown in FIG. **14** and FIG. **18A**. In particular, FIG. **18B-1** depicts portions of two adjacent lens housings **430**, located around a seam **400**, where each housing support an outer compressor lens element **437** and at least a second compressor lens element **438**. The lens housings **430** include internal light traps **457** and sides that extend up into grooves **433** that have been cut into the edges **432** of the outer lens elements **437**. The outer walls of the lens housings **430** are tapered, expanding towards the device center, so as to provide greater mechanical rigidity and robustness. The interaction of two pairs of adjacent channel to channel datums **335** can be seen in cross-section.

[0153] FIG. **18B-2** depicts an alternate cross-sectional view of these same components, but cut along or within a seam **400**, to show the interaction of channel to channel datums **335**. As in FIG. **18A**, the adjacent pair channel to channel datums **335a** and **335b**, each have curved surfaces **373** that interact locally to help register or align adjacent camera channels together. In this example, to further reduce the potential for over-constraint, the channel to channel datum pairs are asymmetrical and partially offset and are thus more likely to provide localized point contacts and less likely to cause over-constraint. It is noted that the circular holes **530** provide access through a side wall of a lens housing **430**, to enable application of adhesive or RTV used in mounting an outer lens element **437** to the housing.

[0154] As an alternative to the camera channels **320** having a ball pivot **340** at the end of a tripod or channel centering hub **330** (e.g., FIG. **13** and FIG. **17**), the tripods can have a concave socket instead. In particular, the socket can have a concave surface that contacts a nominally matching convex surface of a spherical central hub (not shown). The central hub, which is nominally positioned with its center at the device center, and can be attached to a support post, has a series of mounting areas, each of which interfaces with a concave socket of a camera channel. A cable, having a ball at an end, can be used to pull or tension the socket against the central hub. The primary channel, and each of the secondary channels, can be held together by similar tensioned cables that descend into the support post, where they are fastened and locked. Alternately, the primary channel can be held in place with a tightened bolt. As compared to the prior approaches of FIG. **13** and FIG. **17**, this approach has exchanged the plurality of balls and sockets for an inverted configuration with a plurality of sockets (one per camera channel) contacting one main ball or hub. The tensioned cables replace the prior approaches that used retention pins, or latches, or springs. This approach can enable numerous camera channels to be simultaneously and robustly pulled into alignment against the ball hub, and about the device center. The ball hub can be machined from a precision ball bearing.

[0155] During assembly and alignment of the camera channels, it is also important to properly position the lens elements within the housings. For example, the compressor lens (**437**) can be provided with datum features along edges **432** that interact with a set of mating datum features on the inside of the lens housing **430**. The datum features provided on the lens elements are intended to limit perpendicularity and concentricity error and are designed to be reasonably machined or ground features, or to be mounted onto the ground glass lens bevel surfaces with an adhesive. As shown in FIG. **16**, these features can include flat datum surfaces **650** fabricated on the bottom face of the compressor lens at the corner, outside the FOV. Other datums, including adjacent flat edges **660**, that can be ground into the round lens element before it was shaped into a pentagon with truncated or beveled edges **370**, can be mated with flat surface datums **670** on the lens mount. As glass is typically mounted to metals with adhesives, these flat edge datums can provide a guiding alignment without risk of over constraint. The outer lens element **437** can also be provided with an anti-clocking datum feature **680**, likely bonded to one of the beveled surfaces.

[0156] Because of these datum features, the chamfered or beveled finishes on the pentagonal lens

surfaces can be fabricated with significantly less precision, thus helping to reduce lens cost. Even though FIG. 13 depicts a pentagon-shaped compressor lens or outer lens element **437** and housing **430**, these mechanical approaches to reduce mis-alignment errors can also be applied to hexagonally shaped lenses, or lens elements having other polygonal shapes.

[0157] It is recognized that in providing an improved multi-camera panoramic image capture device **300** with outer lens elements **437** having inner beveled edges **370** (FIG. 13), the centering tolerances can be worse as compared to typical lens elements fabricated with conventional cylindrical edges. Such decentering can in turn effect the centering of the FOV captured by the entire camera lens system **320**. As one approach for compensation of such errors, the positioning of the effective image centroid or center pixel can be determined either optically or electronically. As another approach, an improved lens housing **430** can be designed to provide a compensating lateral adjustment for one or more internal lens elements. For example, means to adjust the positioning or tilt of an intermediate inner lens element, such as for a lens element located between the stop and the image sensor, can be provided. The adjustment means can include or use micrometers, pins, shims, flexures, or springs. Z-axis compensators to adjust focus or magnification differences can also be provided within each camera channel by similar mechanisms. The mechanisms to enable compensation can be built in, or internal to the device or a camera, or external, or a combination thereof.

[0158] For example, the axial alignment or focal position of an image sensor, relative to an image plane provided by a camera lens assembly (**320**) can be improved by means of an appropriate mechanism. For example, three adjustment screws can be used to control X translation along with Z-axis rotation, and another set of three screws can be used to control Z-axis translation along with X-Y axes rotations, and an additional screw is used to control Y axis translation. A pair of springs can be used to retain a gimbal plate, while also allowing X and Z-axis translations respectively. Other adjustment designs or devices can be used, such as using pins and micrometers or pins and shims, within the tight space constraints that the camera **320** and the overall panoramic multi-camera capture device **300** allows. An optical fiducial, using a known controlled light source, can also be used within a camera assembly to enable an image centroid pixel on the image sensor to be identified and tracked. Use of an optical fiducial can also aid identification, tracking, and correction of the polygonal edges of the polygonal FOV captured by a camera channel. Similarly, the camera channels can also have baffles built into their lens housings that provide sharp shadows at the image plane or sensor. For example, a camera channel can have a baffle located between an outer lens element **437**, or compressor lens, and subsequent inner lens elements. The baffle can provide a sharp polygonal edged aperture (e.g., pentagonal or hexagonal) following the nominal polygonal shape of the outer lens element **437** and the core FOV **205**, and also a blackened surface for blocking and absorbing light outside the intended FOV. Alternatively, the light absorbing baffle can be painted or coated on an internal lens element surface. Thus, the baffle or mask can also define the edges of the transiting image light and thus cast an edge shadow onto the image sensor **270**. As such, the baffle can help sharply define the illuminated polygonal edges of light incident to the image sensor, and thus assist in the determination and spatial and temporal tracking of an imaged polygonal FOV, for either a core FOV **205** or an extended FOV **215**.

[0159] Alternately, or in addition, to the device and camera assembly approach of FIGS. 13-14 and FIGS. 16-18, the lens housings **430** can be equipped with one or more tab or post like structures (not shown) that can protrude out from a lens housing out of the nominal conical space or volume and interact with similar protruding structures of an adjacent lens housing, or alternately interact with indented structures of an adjacent lens housing. For example, these protruding structures can be provided in the general vicinity of the sensor **270** and sensor package **265**, and be several millimeters in length, and have datum features to help kinematically control a DOF, such as a radial position from the device center, or a tilt or rotation of one camera channel to an adjacent camera channel. A camera assembly can have two such protruding structures, arranged symmetrically or

asymmetrically about the lens housing and one oriented orthogonally to the other, to control a different degree of freedom. Alternately, or in addition, the camera channels can have lens housings **430** that include one or more protruding tab, or post structures located within the seams **400**. For example, such datum structures (not shown) can be provided within the seams, at the polygonal outer face of a camera channel, at a vertex **60**, and can protrude out of the nominal conical space or volume and into the seam **400** between the two adjacent channels. Depending on the device design and intended applications, the protruding tab or structures, whether located in the outer seams **400**, or more deeply imbedded, such as near the image sensors, can be either be fabricated from a compliant or a stiff material, or a combination thereof. As another alternative, one or more tabs of a given lens housing **430** need not protrude outside the nominal conical volume or frustum, but rather clamps that bridge from one tab to a tab of an adjacent lens housing can provide the interface or control to limit a degree of freedom.

[0160] It is noted that in the assembly approach depicted in FIG. **13** and FIG. **17**, and the aforementioned variants and alternatives, in which a primary channel **610**, and secondary channels **615**, having tripods **330** that interface with a central ball socket array or hub, that the available space in the center of the device can be tight. This can make it difficult to design in the power and communications cables, cooling lines, and related support mechanics.

[0161] As another option to provide more access for cabling, supports, and thermal management hardware, the ball and socket approach of FIG. **17** can be replaced with an internal frame (FIG. **19**) with polygonal faces, with access holes to a hollow center. For an improved multi-camera panoramic image capture device **300** constructed in a dodecahedral pattern, the internal frame would also be dodecahedral with pentagonal faces and it would be oriented with the internal pentagonal faces nominally aligned with the external pentagonal geometry. An internal frame can be machined separately and assembled from 2 or more pieces, or it can be made as a single piece structure by casting or 3D printing. Although, the fabrication of a single piece frame could be more complex, the resulting structure can be more rigid and robust, and support tighter mechanical tolerances. For example, a dodecahedral frame with a hollow center could be cast in stainless steel, and then selectively post-casting machined to provide precision datum features. This internal frame can then be provided with flexures or adjustors on all or most of the pentagonal faces, to provide kinematic type adjustments and to reduce or avoid over constraint during device assembly and use. As before, the available adjustors on these internal faces can be different for the secondary channels as compared to a primary channel. Alternately, an internal frame can be at least in part made with a more compliant material, such as brass or Invar. As the central volume of this internal frame can be at least partially hollow, space can then be provided for the electrical cabling, thermal management hardware, and other support structures.

[0162] FIG. **19** provides an example of such an internal frame **800**, with numerous pentagonal faces **810** arranged in a dodecahedral pattern with a hollow center. An internal frame **800** can be designed as a mount mechanical assembly for an 11-camera system, with a support post attaching in the 12th position (similar to FIG. **13** and FIG. **15**). A polygonal internal frame, or half or partial internal frame can also be used in a partial or hemispheric system, where the camera assemblies, including images sensors are mounted to the frame. Alternately, a hemispherical system with an internal frame **800** can use a central hollow space (e.g., a nexus) to enable image light to cross through in reaching image sensors on the far side, including by transiting intervening relay lens systems. As shown in FIG. **19**, a pentagonal face (**810A**) can have three adjustors **820**, such as set screws or flexures, oriented nominally 120° apart, that can interact with features on the camera housing and thus be used to help align a given camera channel. As previously, the mounting and adjustments for secondary channels can have a different design or configuration than those for a primary channel. As another alternative (not shown), one or more pentagonal faces **810A**, **810B**, or **810C** can be provided with one or more adjustors that can be used to nudge the respective camera channel against a precision v-groove structure (also not shown). These-v-groove structures can be

fabricated into, or protruding from, an inside edge of a pentagonal vertex **60** of a pentagonal face. The internal frame approach can be used with other polygonal device structures, such as that for an icosahedron.

[0163] It is noted that in some systems, one or more of the respective camera objective lens systems (e.g., **320** or **920**) can be paired with an image relay optical system to reimage the original image plane to a more distant image plane. The original image plane provided by an objective or camera lens system is essentially a real intermediate image plane within the larger optical system. It can be re-imaged with a magnification (e.g., 1:1 or 2.75:1) to a subsequent image plane (not shown) where an image sensor is located. Thus, advantageously, the image sensor can be larger, and provide a higher pixel count, and the relay lens systems can re-image the image provided by the objective lenses (**920**) at an appropriate magnification to nominally fill the more distant sensor with a projected image. The relay lens systems can include field lenses after the image plane of the camera lenses so as to help contain the space or volume needed for transiting image light. The optics of the relay lens systems can also be coherently designed with the cameras **920**, so as to correct or compensate for aberrations thereof.

[0164] FIG. **20** depicts an alternate optical path in which a low parallax imaging **320** is paired with a relay optical system **725**, that re-images an intermediate or first image plane **755A** to a second image plane **755B**. The relay optics **725** are shown as having all refractive elements (e.g., lens elements), but the relay can also be catoptric (having all reflective elements) or catadioptric (having a combination of refractive and reflective elements). The relay can also be designed to collect image light from an intermediate image plane (image plane **755A**) in either a telecentric or non-telecentric manner, and likewise present image light to the remote image sensor in a telecentric or non-telecentric manner. The relay can also be double telecentric, which means simultaneously telecentric to both the intermediate image and remote sensor planes. Either the aperture stop **945A** or the secondary aperture stop **945B** can be the limiting stop for this system. As an example, the first aperture stop can be the limiting stop, and the secondary aperture stop can be slightly oversized, but can still contribute to vignetting stray light. Image light propagating through a secondary aperture stop may be not be as well controlled, or may be more aberrated, than image light through the primary stop.

[0165] However, it can be difficult to configure an improved low-parallax panoramic multi-camera capture device (**300**) in which the imaging channels further include optical relays, mechanically within a nominally hemispherical volume. The use of prisms or mirrors to fold or redirect the mechanical volume can reduce the overall volume. Alternately, the image light passing into the relay optical systems can cross through each other during transit from an imaging lens **320** and into the respective relay optical system **725**. The internal frame **800** of FIG. **19** can enable this approach, where pentagonal faces **810** on one side of the frame interface with low parallax imaging lenses **310**, and the pentagonal faces on the opposite side interface with the relay system opto-mechanics. The relay optics can have a working distance **760** that nominally matches the diameter of the internal frame **800**.

[0166] As shown in FIG. **20**, the optical relay **725** can also include one or more beamsplitting prisms **735** which can direct light to an image sensor **730** or to other types of sensors. A beamsplitter can be positioned in either telecentric or non-telecentric space, or before or after the aperture stop of the relay optical path. The combination of a beamsplitter and a secondary sensor can be used to support additional or secondary optical functions, such as IR imaging, or light field imaging, or LIDAR depth imaging. A beamsplitting prism can also be used for color splitting to create separate RGB color imaging channels, each with their own image sensor. Then, if lateral color optimization is relaxed in favor of reducing the extent of front color, the resulting image differences can be corrected before the images are digitally recombined during image processing. One or more fold mirrors can also be included within the optical relay.

[0167] As noted previously, the width of the seams **400** can be a critical parameter in the design of

an improved multi-camera capture device **300**, as the seam width affects the parallax error, size of blind regions **165**, the size of any compensating image overlap (FIG. **3**), and the image processing and mosaicing or tiling time. For example, in cases where an expensive multi-camera capture device **300** is used in controlled environments, narrower seams with small tolerances could be allowed or tolerated. However, a local collision or stress of expensive brittle materials (e.g., glass) should be avoided, as chipping and other damage can result. There are of course other external damage risks, including those from external materials coming in contact with the outer lens elements **437**. Thus, as a multi-camera capture device **300** can be an expensive unit, opto-mechanical designs that provide risk mitigation, or protection of the cameras or the device, can be beneficial.

[0168] As one remedy, a seam **400** can be filled with a compliant material, such as an RTV or silicone type adhesive. As another remedy, the optical design of the camera lenses can include an outer lens element **437** that is designed to be a polymer or plastic material, such as Zeonex, that is less brittle than is glass. The combined use of a polymer outer lens element **437** and seams **400** filled with compliant materials can further reduce risks.

[0169] However, in optical designs, glass usually provides better and more predictable performance than do polymer materials. As one approach, the outer lens element **437** may be designed with a common, more robust and less expensive optical glass, such as BK7, although higher index materials will likely be needed for other compressor lens elements. The outer lens elements, whether glass or polymer, can also be overcoated with scratch resistant AR coatings, and oleophobic or hydrophobic coatings. A compliant seam filling adhesive should likewise resist penetration or contamination by water, oils, or other common materials.

[0170] In the improved multi-camera capture device **300**, an entire camera **320** can also be designed to be modular, or a potential field replaceable unit (FRU). In the latter case, the camera **320**, including the housing **430**, and the accompanying lens elements including the outermost compressor lens element **437**, can be removed and replaced as a unit. An improved multi-camera capture device **300**, and the cameras **320** therein, can also be protected by a dome or a shell (not shown) with nominally concentric inner and outer spherical surfaces through which the device can image. The addition of an outer dome can be used to enclose the nearly spherical device of FIG. **15**, or a nearly hemispheric device, or a device having an alternate geometry or total FOV, within an interior volume. The dome can consist of a pair of mating hemispheric or nearly hemispheric domes that interface at a joint, or it can be a single nearly hemispheric shell. The transparent dome or shell material can be glass, plastic or polymer, a hybrid or reinforced polymer material, or a robust optical material like ceramic, sapphire, or Alon. The optically clear dome or shell can help keep out environmental contaminants, and likewise if damaged, function as a FRU and be replaced. It can be easier to replace a FRU dome than an entire camera **320** or a FRU type outer lens element or outer lens element assembly. Depending on the complexity of the design, FRU replacement of modular camera lens assemblies or dome may occur in the field, in a service center, or in the factory, but with relative ease and alacrity.

[0171] The dome or shell can also be enhanced with AR, oleophobic, or hydrophobic coatings on the outer surface, and AR coatings on the inner surface. Although the use of a dome or shell can reduce the need or burden of also using a carrying case or shipping container, such enclosures can still be useful. Alternately, or in addition, the dome or shell can be faceted, and provide a series of integrated adjacent lens elements that function as the outer lens elements for the associated adjacent cameras. This approach can have the potential advantages of reducing the widths of both the intervening seams **160** (e.g., seam widths ≤ 0.5 mm) and their associated blind regions, and enabling the device center **196** to be coincident with the low parallax volumes **188**.

[0172] Preference for narrow widths for the seams **400** has been emphasized. But to clarify, relative to the design and performance of a multi-camera capture device **300**, for chief rays **170** (e.g., rays **970** and **975**) transiting adjacent cameras **320**, and for a given object distance, it can be unnecessary

to hold the beam to beam separation completely to zero millimeters. Rather, a goal can be to have at most a pixel of missing information (or a few pixels, depending on the application) at the object distance of interest. Depending on the allowed seam width and pixel loss, the cameras **320**, and multi-camera capture device **300**, can be more or less robust, or more or less tolerant of both fabrication tolerances and protective design approaches, including those discussed previously. [0173] In particular, some fabrication tolerances of the cameras **320** (e.g., see FIG. **13**), housings **430**, outer lens elements **437**, and seams **400** can be tolerated or accommodated. In part, the use of the optical or electronic fiducials, or a shadow cast by a baffle, enables greater tolerance or flexibility in collecting low parallax error images from adjacent cameras **320** with little FOV overlap, as these corrective approaches can enable a “centered” image to be found. However, a multi-camera capture device **300** can also be tolerant of some image loss. For example, for a 5 foot object distance, and a device whose cameras support an aggregate 32 Megapixel resolution (an 8k output equirectangular image), a 1 pixel width seam **400** can equate to a 1.2 mm sized gap. If an improved multi-camera capture device **300** is designed to hold the mechanical seams to 3 mm after construction, then the cameras can be designed for a 1.5 mm gap while allowing some FOV overlap (□□) The device can have a mechanical design that with tolerances, allows a matching pointing error of at most □□. After the cameras are assembled, the actual rays that are collected parallel to the edge surface that are shared between the camera lenses may shift, but will always be within the pointing error. Thus, a gap or seam **400** can be constrained to be at most 3 mm wide. For some camera designs, and markets or applications, wider seams can be tolerated, in either absolute size (e.g., 4.5 mm seam widths) or in lost image content (e.g., 2-20 pixels per seam). Lost pixels can be compensated for by increasing the FOV overlap or extended FOV **215**, between adjacent cameras, to capture overlapped content but at a cost of some parallax errors and some increase in the image processing burden. Although, for modest amounts of extended FOV (e.g., ≤5%), the residual parallax errors (e.g., FIG. **8B**) can still be modest. The smaller the seams **400**, and the better the knowledge of the position of an image centroid **480**, and an image size and shape, then the larger the core FOV **205** can be on the sensor, and the less lens capacity and sensor area can then be devoted to providing a yet larger extended FOV **215**.

[0174] The seams **400** can be nominally identically wide at the interfaces between abutting adjacent cameras for all cameras **320** in an improved multi-camera capture device **300**. Careful camera and camera housing design, and camera to camera mounting (FIGS. **13** and **14**) can help this to occur. But nonetheless, the seam widths can vary, either upon device assembly, or during dynamic environmental conditions. For example, on one side of a first camera, a seam width between that camera and an adjacent one can be only 0.75 pixels wide, while simultaneously a seam width between the first camera and another adjacent camera can be 3.25 pixels wide. The seam widths can also vary non-uniformly. The resulting images to the sensors can be shifted or rotated relative to expectations. Thus, such variations can increase the parallax errors for the images captured by these adjacent cameras, and complicate image mosaicing or tiling. However, with the use of electronic or optical fiducials, or shadows cast by internal baffles, an image centroid and image edges can be monitored for each camera to enable a quick reference to the nominally calibrated or expected conditions. This reference or correction data can also be compared from one camera to another, with a goal to define effective image centroids and image edges for each camera that effectively best reduces parallax errors for all cameras, either individually, or in aggregate (e.g., average). This information can then be used during image processing to quickly and robustly determine image edges and enable efficient image mosaicing or tiling.

[0175] FIG. **15** depicts an electronics system diagram for an improved multi-camera capture device **300**. In this example, a dodecahedron type device has 11 cameras **320**, and an electro-mechanical interface in the twelfth camera position. Image data can be collected from each of the 11 cameras, and directed through an interface input-output module, through a cable or bundle of cables, to a portable computer that can provide image processing, including live image cropping and mosaicing

or tiling, as well as camera and device control. The output image data can be directed to an image display, a VR headset, or to further computers, located locally or remotely. Electrical power and cooling can also be provided as needed.

[0176] Also, as suggested previously, the performance of a multi-camera capture device, relative to both the opto-mechanics and image quality, can be affected by both internal or external environmental factors. Each of the image sensors **270**, and the entirety of the sensor package **265**, with the data interface and power support electronics can be localized heat sources. To reduce the thermal impact on the camera lenses, and the images they provide, the mechanical design for an improved multi-camera capture device **300** can isolate the sensors **270** thermally from the lens opto-mechanics. To further help reduce thermal gradients between the sensors and their electronics, and the optics, micro-heat pipes or Peltier devices can be used to cool the sensors and re-direct the heat. The heat can be removed from the overall device by either active or passive cooling provided through the electro-mechanical interface in the twelfth camera position, shown in FIG. **15**. This cooling can be provided by convection or conduction (including liquid cooling) or a combination thereof.

[0177] As also suggested previously, outside ambient or environmental factors can also affect performance of a multi-camera capture device. These factors can include the effects of the illuminating ambient light, or thermal extremes or changes in the environment. For example, as sun light is typically highly directional, a scenario with outdoor image capture can result in the cameras on one side of the device being brightly illuminated, while other cameras seeing plenoptic illumination from the scene are in shadows. In such instances, the captured images can show dramatic exposure variations, which can then be modified by exposure correction, which can be provided locally (see FIG. **15**). In an improved multi-camera capture device **300**, light from an optical fiducial can also be used for exposure correction of captured images. Light or pixel signals from a portion of the peripheral image region, between the edges of the core FOV **205** and the extended FOV **215**, can also be used for exposure correction, prior to the image being cropped down to the size of the real current core FOV **205**. It is also noted that as extended FOV **215** of a first camera can overlap at least in part with an extended FOV **215** of an adjacent camera, that light level and color comparisons can be made for content or signals that are simultaneously captured by both cameras. The signals or pixel data from these overlapping regions can be used to determine exposure variations between the two cameras, by having a common reference point (e.g., a matched feature point—using SIFT, SURF or a similar algorithm for finding common feature points in the overlap region).

[0178] It is noted that the peripheral image or exposure data can also be retained for use later in post image processing. Additionally, exposure correction can also be enabled by imbedding optical detectors in the seams **400**, or vertices, between outer lens elements **437**. These abrupt exposure differences can also cause spatial and temporal differences in the thermal loading of some image sensors **270**, as compared to others, within a multi-camera capture device **300**. The previously discussed sensor cooling, whether enabled by heat pipes, heat sinks, liquid cooling, or other means, can be designed to account for such differences. The performance can be validated by finite element analysis (FEA).

[0179] Alternately, one or more camera systems can be protected by the attachment of a shield or mask to cover the polygonal shape, from seam to seam, and vertex to vertex, of the outer lens element thereof. Such shields can be provided to cover a single camera lens system, or multiple systems. These shields can be shaped to generally conform to the outer surface shape of the outer lens elements, and they can be used to prevent saturated exposure or overexposure from bright directional light (e.g., solar), or to prevent contamination from a localized directional environmental source. While these caps are nominally detachable, for some user applications, they may remain in use for a prolonged time period. Overly bright exposure, from the sun or other light sources can also be controlled with an image sensor having electronic shuttering or drains, or a

physical shutter or an electro-optical neutral density, photochromic, or electrochromic filter, that can, for example be designed into a camera **320**, within for example the grouping of inner lens elements **440**. Signals to initiate or control electronic or secondary shuttering can be obtained from the image sensor or from other internal or external optical detectors. As another robustness improvement, one or more camera channels can use dichroic color filter arrays integrated into the image sensor package instead of the standard dye-based CFAs.

[0180] Environmental influences can also cause a multi-camera capture device to be heated or cooled asymmetrically. The previously discussed kinematic mounting or linkage of adjacent camera housings **430** for an improved multi-camera capture device **300** can help reduce this impact, by trying to deflect or average mechanical stresses and limit mechanical displacements. However, it can be additionally beneficial to provide channels or materials to communicate or shift an asymmetrical thermal load to be shared more evenly between or by cameras **320** and their housings **430**. With respect to FIG. **13**, this can mean that the spaces around the lens housing **430** and the channel centering hub **330**, such as the inner volume **390**, can be at least partially filled with compliant but high thermal contact, thermally conductive materials (e.g., Sil-Pad (from Henkel Corporation) or CoolTherm (Lord Corporation, Cary NC)) to help spatially average an asymmetrical thermal load or difference. Alternately, or additionally, thermal conductive straps or tapes, such as an adhesive tape in the 88xx series from 3M (St. Paul, MN) can be used. However, at the same time, some of the effect of thermal changes, relative to the imaging performance of the camera lenses **320**, can be mitigated by both judicious selection of optical glasses and athermal mounting of the optical elements within the lens housing **430**. Taken in combination, an effective design approach can be to enable thermal communication or crosstalk between lenses **320** and their housings **430** to environmental influences, but to simultaneously isolate the lenses and housings from the sensors **270** and their electronics.

[0181] An improved camera **320** for use in an improved multi-camera image capture device **300** can also use a tunable lens element to correct for thermally or mechanically induced focus changes (e.g., defocus). This tunable lens can preferentially be located amongst the inner lens elements **440**, and can be a liquid crystal or elastic polymer type device, such as an electrically actuated focus tunable lens from Optotune (Dietikon, SW).

[0182] The emphasis has been on developing improved cameras **320**, that have a polygonal shaped outer lens element and capture and image light from a polygonal FOV, for use in an improved multi-camera image capture device **300**. Multiple such adjacent cameras can be used in a nominally spherical device or hemispherical device. However, devices **300** can be developed that have a yet smaller number of cameras and that cover a yet smaller total FOV. For example, systems with only four or six low-parallax, adjacent polygonal cameras can be suitable for some market applications. Additionally, a single camera having a polygonal shaped outer lens element that captures image light from a nominally matching polygonal shaped FOV can be use in isolation, such as for security or surveillance applications. For example, a single camera, optomechanically designed to fit within the form of $\frac{1}{8}$ sup.th of an octahedron can be mounted in a ceiling corner of a room, and capture image content of the room environment with little or no blind regions. For a camera for such an application, the low-parallax optimization can be different, as there is not a concern with abut adjacent low-parallax lens systems. Likewise, the emphasis has been on the development of improved camera lens systems **320** for which parallax errors can be reduced within at least a core FOV **205**, while modest extended FOVs **215** (e.g., $\leq 5\%$ extra) and image capture overlap with adjacent cameras can also be provided. Similarly, it is noted that the present approach can be extended to support possible applications for which having yet larger overlapping FOVs (e.g., 10-25% extra, or $\approx 4-10^\circ$ extra FOV for dodecahedral systems with a nominal Core FOV of 37.45°) of image capture between adjacent cameras, and simultaneous parallax error or perspective error control within at least a core FOV, is a priority. The camera designs can be extended even further, to provide yet larger overlapping FOVs (e.g., $10-20^\circ$ extra), but without benefit of reduce parallax

for angles much beyond the designed core FOV.

[0183] As noted previously, the present approach can be used to design low parallax lens systems for a variety of applications. A method of device development can be outlined, for example, as follows:

[0184] 1. Define for a customer or an application space, a polygonal device configuration, a resolution, an expected extent of the seams and blind regions, and a minimum object distance. The minimum object distance is the closest distance at which instant image stitching or image combining of low parallax overlapped images can be applied if image processing is acceptable. At this distance, estimate a maximum total gap or seam width for which the image losses or differences are generally imperceptible to a human viewer. For example, for an 8k equirectangular projected image, an expected image loss of 1/40th % of the output image (along the equator) equates to seams that are 2 pixels wide each. An alternate minimum object distance, for which even modest image processing is not acceptable, and which is further away, can be defined.

[0185] 2. Design the imaging lenses, while controlling both parallax and front color. Depending on the results, the design can be further modified to provide an extra or extended FOV in which the parallax can be controlled. This extra FOV can also provide margin for the rainbow tinting of the residual front color to be located outside of the Core FOV with some margin.

[0186] 3. Design the lens and device opto-mechanics. Determine an expected seam width and the expected thickness and wedge variations between and within the seams, from fabrication and assembly tolerance variations. Determine an expected extended FOV overlap to cover the expected mechanical and optical seam widths to be at least as high or greater than the maximum expected opto-mechanical wedge and thickness variations so that parallel chief rays between adjacent lens systems can be selected without underlap (fields of view that diverge).

[0187] 4. Update the lens design to provide field of view overlap along the seams between the outer compressor lens elements, to exceed both the maximum expected mechanical variations and residual front color, while controlling parallax therein.

[0188] In conclusion, a goal of present application is to provide means for designing low parallax lens systems that can have outer (polygonal) shaped lens elements, which can then provide reduced optical seam widths and blind regions when used advantageously with appropriate mechanics that enable mounting of adjacent camera assemblies. In order to push chief rays to the edge of a polygonal surface, aberrations of the entrance pupil, and particularly spherical aberration and axial chromatic aberration of the pupil, should be optimized or reduced. For context, the entrance pupil and exit pupil, which are projected images of the aperture stop in object space and image space respectively, are usually academic curiosities, representing the virtual images of the aperture stop that a person can see when looking into a lens.

[0189] In a typical optical system, to provide good image quality, aberrations matter at an image plane, with the typical goal to have small net sums of the individual surface contributions, even though values at individual internal surfaces, whether positive or negative, can be magnitudes larger. Whereas, aberrations at the aperture stop often do not matter much, other than positioning the stop correctly and defining the lens system f-number while minimizing clipping or vignetting of the ray bundles versus field. It is noted that if an object was located at the aperture stop, pupil aberrations would affect the apparent image quality of the image of that object as seen by the person viewing the entrance or exit pupil, although these pupil aberrations are not necessarily representative of the image quality at the image plane.

[0190] In low-parallax lenses, on the other hand, pupil aberrations, and particularly entrance pupil aberrations, matter. First, to begin to properly design the lens to control parallax, the entrance pupil must be positioned behind the image plane. Secondly, the aiming of peripheral chief rays from the outer compressor lens element, and towards the low parallax volume, must be controlled. As noted previously, optimizing spherical aberration of the entrance pupil can be an effective way to limit parallax errors. In that context, modest over-corrected or under-corrected optimizing spherical

aberration of the entrance pupil can occur, meaning that the non-paraxial chief ray NP points can lead or follow the paraxial NP point respectively. Additionally, axial chromatic aberration of the entrance pupil creates color differences that can affect optimization of spherical aberration of the entrance pupil. Front color, which can be considered to be an artifact of this axial chromatic aberration, and the axial chromatic aberration itself, can be reduced by judicious choice and use of high and low dispersion optical materials within the lens design. Although optimization of the spherical aberration of the entrance pupil or chief ray pointing provides fine control for the non-paraxial, chief ray NP point position and parallax reduction, distortion from the compressor lens group also has an increasing influence, versus increasing field, on the projected chief ray pointing towards a location behind the lens and towards the low parallax volume. The magnitude and characteristics of the distortion, which also defines the chief ray height on the outer surface of the outer compressor lens element, can be significantly determined by the use of aspheric surfaces within the compressor lens group.

[0191] Although this discussion has emphasized the design of improved multi-camera image capture devices **300** for use in broadband visible light, or human perceivable applications, these devices can also be designed for narrowband visible applications (modified using spectral filters, ultraviolet (UV), or infrared (IR) optical imaging applications. Polarizers or polarizer arrays can also be used. Additionally, although the imaging cameras **320** have been described as using all refractive designs, the optical designs can also be reflective, or catadioptric and use a combination of refractive and reflective optical elements.

Claims

1. An imaging device comprising: a first imaging lens configured to image a portion of incident light from within a first field of view; and a second imaging lens configured to image a portion of incident light from within a second field of view adjacent to the first field of view, the first imaging lens comprising: a first lens element group proximate an exterior surface of the imaging device, the first lens element group having at least an outer lens element that collects incident light in the first field of view to provide image light; a second lens element group configured to receive the image light from the first lens element group; an aperture stop configured to receive the image light from the second lens element group; and a third lens element group configured to redirect the image light from the aperture stop to form an image at an image plane, wherein: first projections of paraxial chief rays included in the incident light that enters the outer lens element converge to an entrance pupil located behind the image plane, second projections of non-paraxial chief rays included in the incident light that enters the outer lens element converge to a low parallax point, and a distance between the entrance pupil and the low parallax point reduces parallax errors for non-paraxial chief rays received proximate an edge of the outer lens element.
